

UPCAT Reviewer

Arranged by:

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As you take this, have pieces of paper and pen/pencil with you.

Practice lang ng practice 😊

Dear future scholar,

*While I was in my undergraduate years, nagpapart-time tutor ako ng mga graduating SHS para sa pagrereview nila sa CETs and fortunately nalalampasan naman nila. Ngayon na may full time work na ako as a Physics Teacher, hindi ko na magagawang maglaan ng oras para magpart time ng reviews. Perhaps napag-isipan ko ito bilang paraan sa pagtulong sa inyong nagrereview until naging advocacy ko na ang tumulong sa mga kagaya niyo. **So, claim this copy for FREE.***

Ngayon, may simple request lang sana ako. Subscribe ka sa youtube channel ko.

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Nag-uupload ako ng academic tutorial videos dito, mock exams, and I can even upload things like tips on studying CETs, etc. Kapag sa kalagitnaan ng pag-aaral mo ay may request kang tutorial na ipa-upload, sabihin mo lang sa akin through comments or message/email.

Join ka na rin sa facebook group na ito for updates and exchanging of ideas

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And finally, ang bagay lang na maipapangako ko ay tutulungan kita sa makakaya ko. So feel free to message/email me for questions.

Magtiwala ka sa sarili mo, and make the ones you love proud!

UPCAT Complete Reviewer

Arranged by: Jazzver O. Sanchez

Detailed solutions for Math are solved by: JAZZVER O. SANCHEZ

For any questions or clarifications, you may send an email at

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I made this FREE for everyone. Hoping that you will pass the exam of your university choice. Have the very best of luck 😊

Note that this is just a reviewer and cannot guarantee of passing the UPCAT. **The passages used in this mock-exam do not necessarily reflect my personal opinions. This is a compilation of materials gathered from various books and electronic sources. I don't claim copyright on materials and examples taken from references.** For the mathematics part, I am going to provide the detailed solutions at the end of the page as well as its complete video tutorials.

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These inspirational quotes might boost your mood in taking this mock-exam.

"Success is not final; failure is not fatal; it is the courage to continue that counts"

"Allow yourself to be a beginner. No one starts off being excellent."

"Push yourself because no one else is going to do it for you."

"Be one of those rare people who step over their insecurities and succeed."

"FEAR – Forget Everything and Run ... or ... Face Everything and Rise. It's your choice."

"Behind every unsuccessful person, there are a lot of unsuccessful years."

"The secret to getting ahead is getting started."

"A year from now you'll wish you had STARTED TODAY."

"A little progress each day adds up to big results."

"Don't let yesterday take up too much of today."

"If you don't sacrifice for what you want, what you want becomes the sacrifice."

"STUDY while others are sleeping; WORK while others are loafing; PREPARE while others are playing; DREAM while others are wishing"

"A journey of thousand miles begins with a single step."

"There is no elevator to success, you have to take the stairs."

"Success doesn't come from what you do OCCASSIONALLY, it comes from what you do CONSISTENTLY."

"Nothing will work unless you do."

"Mistakes are proof that you are trying."

"Yes, though I walk through the valley of the shadow of death, I will fear no evil: for you are with me. (Psalm 23:4)"

"I can do everything through Christ who gives me strength. (Philippians 4:13)"

I am wishing all the best for you. Good luck and God Bless 😊

LANGUAGE PROFICIENCY

Error Identification. These sentences will test your ability to recognize grammar or usage errors. There are three/four underlined portions that may contain errors and a fourth/fifth portion which indicates that the phrase has no error. Select the letter of the underlined portion that has an error. There is no more than one error in every item.

1. The new government was drafting a constitution, electing a legislative body and established diplomatic relations. No error.

A
B

C
D
2. Neither Senate or Congress wants to pass the bill suggested by the President. No error.

A
B
C

D
3. He went into his room, closed the door, opened the window and goes to bed. No error.

A
B
C

D
4. When you pass the UPCAT, one should be proud. No error.

A
B
C
D
5. Everyone in our family except my brother and I has eaten lunch. No error.

A
B
C
D
E
6. Hearing the news the whole group went to the province. No error.

A
B
C
D
E
7. Science refer to that dynamic, cumulative systems of verifiable concepts and processes which seek to understand natural phenomena. No error.

A
B
C

D
E
8. The resuscitation of multiple victims of the tsunamis have kept the death toll to a minimum. No error.

A
B

C
D

9. None of the players in the locker room were able to comprehend or to solved the problem. No error.
A B C
D E

10. A number of optional sections have been included. No error.
A B C D E

11. To enjoy all rewards and privileges, please present these card when purchasing at the store. No error.
A B C D
E

12. Sitting along the beach across the clinic, Lito wondered what he was suppose to do. No error.
A B C D
E

13. I guess he sat down resting in the shade and never got up again. No error.
A B C D E

14. Some scientists consider Sergey Koryolev's rockets to be more revolutionary than Von Braun of NASA. No error.
A B
C D

15. As passionate students, Jelay and May plan to become a passer of the upcoming UPCAT. No error.
A B C
D

16. Natatakpan ng kadiliman ang iba't-ibang makukulay na larawan. Walang mali.
A B C D

17. Napag-alaman na nakipagsabwatan siya sa kanyang tauhan upang maikubli ang katiwaliang kanyang ginawa. Walang mali.
A B C
D

18. When I called you up this morning, what are you doing? No error.

A

B

C

D

E

19. As dedicated teachers, Kate and Sophia plans to become members of the

A

B

volunteer teaching team. No error.

C

D

20. Some of the boys who plays basketball also like to play football. No error.

A

B

C

D

21. Hindi bumoboto doon si manong Jose dahil mahaba ang pila. Walang mali.

A

B

C

D

22. Kundi man ngayon makaluwas si Jorge sa Maynila, ay sa Agosto pa siya

A

B

luluwas. Walang mali.

C

D

23. He was quite amusing when he heard what had happened. No error.

A

B

C

D

E

24. He stopped to see if he could picked up the trail. No error.

A

B

C

D

E

25. Either Lisa or Karen will always volunteer their valuable time to serve on our

A

B

C

D

board. No error.

E

26. Work as quick as you can but as carefully as possible when you take the test. No

A

B

C

D

error.

E

27. Professor Lane, our Computer Science teacher, was excited when he had the

A

B

opportunity to meet Bill Gates, the president of Microsoft, Inc. No error.

C

D

E

- a.) dala-dala
- b.) natatanaw

- c.) problema
- d.) kasalanan

36. The book Crime and Punishment is considered by many to be Dostoevsky's magnus opus.

- a.) greatest work
- b.) favorite

- c.) longest work
- d.) achievement

37. Communism is considered as the antithesis of capitalism.

- a.) opponent
- b.) opposite

- c.) raconteur
- d.) depredation

38. No person could articulate his ideas better than the council president.

- a.) say
- b.) present

- c.) organize
- d.) translate

39. Much of his writing is slack that is why readers get bored at his works.

- a.) lacks action
- b.) lacks completeness/perfection

- c.) lacks firmness
- d.) lacks diligence

40. The gases trapped in pasty lava do not escape freely and may sometimes break out with explosive violence.

- a.) sudden
- b.) expressive

- c.) unrestrained
- d.) volatile

41. In the U.K. the name of the Scottish city Edinburgh, in the vernacular, is actually pronounced as "E-din-burrah"

- a.) Native Language or Dialect
- b.) Local Slang

- c.) Peculiar manner
- d.) Surrounding area

42. The Apotheosis of Washington is a unique work of art that can be found in the U.S. Capitol.

- a.) Glorification b.) Deification c.) Inauguration d.) Archetype

43. Leo made a disappointed, misunderstood face, from the many creases at his disposal.

- a.) authority b.) distribution c.) garbage unit d.) regulation

44. The nomadic Mongols lived in yurts since the days of Genghis Khan.

- a.) Tents b.) Houses c.) Caravans d.) Wagons

45. The defendant was acquitted by the court in yesterday's hearing.

- a.) arrested b.) absolved c.) convicted d.) released

46. Joanna's reply to his query was sure and crisp, that it didn't took too long before she was able to prod him away.

- a.) brittle b.) lively c.) concise d.) fragile

47. Artifice was one of the qualities of style that was worthy of further cultivation.

- a.) fostering of growth c.) development
b.) refinement d.) elevation

48. Biscuits had been stamped on the floor, making mopping more laborious than it already is.

- a.) crushed b.) impressed c.) pounded d.) imprinted

49. Please depress the red button when you're ready.

- a.) press down b.) repress c.) discourage d.) weaken

50. Naniniktik ang mga Espanyol sa mga posisyon ng mga Katipunero.

- a.) Umaatake b.) Nanghihimasok c.) Nag-eespiya d.) Sumusugod

51. Hindi napukaw ang atensyon ng mga manonood ng bagong kanta.
- a.) nakuha b.) naantala c.) nasik d.) nagulat
52. Si Salvi ang may pasimuno ng lahat ng ito.
- a.) tumigil b.) nagsimula c.) nagpakalat d.) naghasik
53. The shaft that kept the roof up stayed rigid, even as the wind and rain gained strength.
- a.) Pliable b.) Flaccid c.) Firm d.) Flexible
54. Back in her time, Margaret Thatcher was an eminent world leader, although she was very unpopular with the U.K.'s leftist north.
- a.) famous b.) obscure c.) unimportant d.) articulate
55. In medieval England, papists, or Catholics, could be obtruncated if caught.
- a.) deported b.) beheaded c.) tortured d.) imprisoned
56. George Orwell brilliantly portrayed a socialist dystopia in his book 1984.
- a.) Fable b.) Anti-utopia c.) Myth d.) Utopia
57. As a result of inbreeding, some members of the Russian Imperial household suffered from hemophilia.
- a.) Bloodthirstiness c.) Uncontrollable Bleeding
b.) Infertility d.) Mental Disorders
58. One week later, the vessel deposited them in one of the biggest ports on the Eastern Peninsula.
- a.) consigned b.) dropped c.) credit d.) accumulated
59. Madalas kang makakakita ng sambalilo sa ulo ng kinakalbong si Padre Francisco.

- a.) Sombrero
- b.) Kapa

- c.) Huwad na buhok
- d.) Takip sa ulo ng mga pari

60. By the end of the century, mass circulation of newspapers had been developed.

- a.) movement of blood through veins and capillaries
- b.) dissemination
- c.) distribution of copies
- d.) availability in libraries

SENTENCE COMPLETION. Select the word or phrase that most appropriately completes the sentence.

61. The police officer _____ the crowd to step back from the fire so that no one would get hurt.

- a.) undulated
- b.) enjoined
- c.) stagnated
- d.) permeated

62. Bea was known for her loud and domineering personality and was considered a _____ by many who knew her.

- a.) banality
- b.) debutante
- c.) virago
- d.) trifle

63. Nothing will _____ my memory of the night we first met; the images are forever burned in my mind.

- a.) appease
- b.) undulate
- c.) inculcate
- d.) efface

64. Sheila's grueling hike included passing through numerous _____.

- a.) terrariums
- b.) neoprene
- c.) jurisdictions
- d.) ravines

65. The artist attempted to _____ the painting by adding people dressed in bright colors in the foreground.

- a.) excoriate
- b.) amplify
- c.) eradicate
- d.) vivify

66. The _____ deer stuck close to its mother when venturing out into the open field.

- a.) starling
- b.) foundling
- c.) yearling
- d.) begrudging

67. David's _____ entrance on stage disrupted the scene and caused the actors to flub their lines.
- a.) untimely b.) precise c.) lithe d.) fortuitous
68. Calvin reached the _____ of his career in his early thirties when he became president and CEO of a software company.
- a.) zephyr b.) zenith c.) vale d.) nocturne
69. Genevieve's stunning debut performance at the city opera has earned her _____ from some of the city's toughest critics.
- a.) antipathy b.) accolades c.) destitutions d.) lamentations
70. Prince Phillip had to choose: marry the woman he loved and _____ his right to the throne, or marry Fiona and inherit the crown.
- a.) reprimand b.) upbraid c.) abdicate d.) winnow
71. Ming's blatant lie revealed that he suffered no _____ about being dishonest to his parents.
- a.) compunction b.) repudiation c.) vindication d.) evanescence
72. The log cabin had a(n) _____ feel, so it was often enjoyed by vacationers from the city throughout the year.
- a.) abstract b.) ersatz c.) rustic d.) raucous
73. For _____ deeds during her mission overseas, Tyesha was awarded the Congressional Medal of Honor.
- a.) inept b.) valorous c.) erroneous d.) malodorous
74. Sunlight shining through a window was an obvious _____ in the nearly every one of the artist's works.
- a.) disjunction b.) hindrance c.) repugnance d.) motif

75. Several weeks of extremely hot, dry weather _____ the land, so instead of rowing across a river, we walked across a cracked, parched riverbed.
- a.) oscillated b.) desiccated c.) subverted d.) coalesced
76. As the pressures of her business became overwhelming, Charlotte chose to _____ her role as PTA president.
- a.) expedite b.) transgress c.) propagate d.) relinquish
77. The heat was absolutely _____, making everyone irritable, sweaty, and uncomfortable.
- a.) oppressive b.) salient c.) replete d.) prosaic
78. When the house on the corner burned down, the entire neighborhood _____ together to help to the victims re-establish their lives.
- a.) rallied b.) recited c.) skulked d.) disintegrated
79. Frank feels such _____ towards his ex-business partner that he cannot stand to be in the same room with him.
- a.) iniquity b.) collusion c.) animosity d.) pallor
80. Oliver was unable to _____ himself from the difficulties he had caused by forging the documents.
- a.) reprove b.) pique c.) oust d.) extricate
81. When Melinda arrived in the impoverished city, she was immediately _____ by bands of children begging for food.
- a.) bedraggled b.) accosted c.) infiltrated d.) rebuked
82. Given his _____ nature, it was appropriate that he decided to be a trial lawyer after law school.
- a.) lackluster b.) engrossed c. penitent d. litigious

83. Buying homeowner's insurance is a wise decision because it provides _____ if your house should be damaged in a fire.

- a.) chastisement b.) indemnity c.) clarification d.) resolution

84. Tabitha found an overpriced dining table at the antique shop and tried to _____ with the shopkeeper, but he refused to lower the price.

- a.) haggle b.) stipulate c.) announce d.) ruminate

85. With Justine's _____ nature and passion for art, she would make an excellent tour guide for the museum.

- a.) volatile b.) congenial c.) servile d.) fledgling

READING COMPREHENSION

READING COMPREHENSION. Read the given passage and answer the following questions.

The victory of the small Greek democracy of Athens over the mighty Persian Empire in 490 B.C. is one of the most famous events in history. Darius, king of the Persian Empire, was furious because Athens had interceded for the other Greek city-states in revolt against Persian domination. In anger the king sent an enormous army to defeat Athens. He thought it would take drastic steps to pacify the rebellious part of the empire.

Persia was ruled by one man. In Athens, however, all citizens helped to rule. Ennobled by this participation, Athenians were prepared to die for their city-state. Perhaps this was the secret of the remarkable victory at Marathon, which freed them from Persian rule. On their way to Marathon, the Persians tried to fool some Greek city-states by claiming to have come in peace. The frightened citizens of Delos refused to believe this. Not wanting to abet the conquest of Greece, they fled from their city and did not return until the Persians had left. They were wise, for the Persians next conquered the city of Eritrea and captured its people.

Tiny Athens stood alone against Persia. The Athenian people went to their sanctuaries. There they prayed for deliverance. They asked their gods to expedite their victory. The Athenians refurbished their weapons and moved to the plain of Marathon, where their little band would meet the Persians. At the last moment, soldiers from Plataea reinforced the Athenian troops.

The Athenian army attacked, and Greek citizens fought bravely. The power of the mighty Persians was offset by the love that the Athenians had for their city. Athenians defeated the Persians in both archery and hand combat. Greek soldiers seized Persian ships and burned them, and the Persians fled in terror. Herodotus, a famous historian, reports that 6,400 Persians died, compared to only 192 Athenians.

86. Athens had _____ the other Greek city-states against the Persians.

- | | |
|-----------------------------|-----------------------------------|
| a.) refused help to | c.) wanted to fight |
| b.) intervened on behalf of | d.) given orders for all to fight |

87. Darius took drastic steps to _____ the rebellious Athenian.

- | | | | |
|------------|-------------|----------|--------------|
| a.) weaken | b.) destroy | c.) calm | d.) irritate |
|------------|-------------|----------|--------------|

88. Their participation _____ to the Athenians.

- a.) gave comfort b.) gave honor c.) gave strength d.) gave fear

89. The people of Delos did not want to _____ the conquest of Greece.

- a.) end b.) encourage c.) think about d.) daydream about

90. The Athenians were _____ by some soldiers who arrived from Plataea.

- a.) welcomed b.) strengthened c.) held d.) captured

Desertification, the creation of desert-like conditions where none had existed before, is the result of the vagaries of weather and climate or the mismanagement of the land or, in most cases, some combination of both. Such ecological deterioration in the Sahel has been linked in several ways to the increased size of livestock herds. During the fifteen years preceding 1968, a period of extremely favorable rainfall, the pastoralists moved into the marginal regions in the north with relatively large herds. However, with the onset of a series of dry years beginning at the end of the rainy season in 1967, the pastoral populations found themselves overtaxing very marginal rangelands, with the result that the nomads viewed themselves as victims of a natural disaster. The mistaken idea that drought is an unexpected event has often been used to excuse the fact that long-range planning has failed to take rainfall variability into account. People blame the climate for agricultural failures in semiarid regions and make it a scapegoat for faulty population and agricultural policies.

Deterioration and ultimately desertification in the Sahel and in other ecosystems can be combated only if an ecologically realistic carrying capacity for the rangelands is determined. Although there appears to be widespread agreement that such a determination would be significant, there has been little agreement on how to make operational the concept of carrying capacity, defined as the amount of grazing stock that the pasture can support without deterioration of either the pasture or the stock. Should the carrying capacity be geared to the best, the average, or the poorest years? Which combination of statistical measures would be most meaningful for the planning of long-term development of rangelands? On which variables should such an assessment be based, vegetation, rainfall, soil, ground and surface water, or managerial capabilities? Such inconclusiveness within the scientific community, while understandable, creates confusion for the land managers, who often decide to take no action or who decide that all scientific suggestions are of equal weight

and, therefore, indiscriminately choose any one of those suggested. Given the downward spiral of land deterioration, it becomes essential that an ecologically acceptable carrying capacity be established and enforced.

It will also be crucial that land managers know what statistical and quasi-statistical measures actually mean: no single number can adequately describe the climate regime of an arid or semiarid region. Land managers must supplement such terms as the “mean” with more informative statistical measures to characterize adequately the variability of the climate. The understanding of this high degree of variability will serve to remove one of the major obstacles to resolving the perennial problems of the Sahel and of other arid or semiarid regions.

91. The author is primarily concerned with _____.

- a.) criticizing a social attitude
- b.) suggesting an approach to solving a problem
- c.) explaining the mechanics of a process
- d.) defending the theories of ecological scientists

92. According to the passage, which of the following contributed to the desertification of the Sahel?

- I. The size of the livestock herds grazing on the land
- II. The quality of the land in the Sahel
- III. The amount of rainfall after 1967

- a.) I only
- b.) II only
- c.) I and III only
- d.) I, II, and III

93. It can be inferred from the passage that the nomadic tribes who moved into the marginal regions of the Sahel did NOT _____.

- a.) enlarge the size of their livestock herds
- b.) conserve water after the drought began
- c.) live in the Sahel after 1968
- d.) expect a drastic change in weather conditions

94. It can be inferred from the passage that the concept of the carrying capacity of land is _____.

- a.) still hypothetical rather than practical
- b.) basically political rather than ecological

- c.) independent of climatic conditions
- d.) relatively unknown among ecologists

95. Which of the following best states the author's view concerning the relationship between the ecological scientist and the land manager?

- a.) The scientist has not provided the manager with clear guidelines that can be used in regulating the productivity of land.
- b.) The scientist has provided theories that are too detailed for the manager to use successfully.
- c.) The scientist and the manager, in attempting to regulate the use of semiarid land, have ignored the traditional behavior patterns of pastoral communities.
- d.) The manager has misunderstood and hence misapplied the suggestions of the scientist.

96. With which of the following statements concerning desertification would the author be most likely to agree?

- a.) It is the result of factors beyond the control of science.
- b.) It is a problem largely affecting arid regions.
- c.) It could be prevented if land managers understood statistics.
- d.) It is not always the result of drastic climate changes alone.

97. According to the passage, a statistical description of the climate regime of an arid or semiarid region would probably be _____.

- a.) misleading
- b.) impossible
- c.) complex
- d.) meaningless

98. The tone of the passage can best be described as

- a.) flippant
- b.) objective
- c.) aggressive
- d.) apologetic

Coral reefs are one of the most fragile, biologically complex, and diverse marine ecosystems on Earth. This ecosystem is one of the fascinating paradoxes of the biosphere: how do clear, and thus nutrient-poor, waters support such prolific and productive communities? Part of the answer lies within the tissues of the corals themselves. Symbiotic cells of algae known as zooxanthellae carry out photosynthesis using the metabolic wastes of the coral thereby producing food for themselves, for their corals, hosts, and even for other members of the reef community. This symbiotic process allows organisms in the reef community to use sparse nutrient resources efficiently.

Unfortunately for coral reefs, however, a variety of human activities are causing worldwide degradation of shallow marine habitats by adding nutrients to the water. Agriculture, slash-and-burn land clearing, sewage disposal and manufacturing that creates waste by-products all increase nutrient loads in these waters. Typical symptoms of reef decline are destabilized herbivore populations and an increasing abundance of algae and filter-feeding animals. Declines in reef communities are consistent with observations that nutrient input is increasing in direct proportion to growing human populations, thereby threatening reef communities sensitive to subtle changes in nutrient input to their waters.

99. The passage is primarily concerned with

- a.) describing the effects of human activities on algae in coral reefs
- b.) explaining how human activities are posing a threat to coral reef communities
- c.) discussing the process by which coral reefs deteriorate in nutrient-poor waters
- d.) explaining how coral reefs produce food for themselves

100. The passage suggests which of the following about coral reef communities?

- a.) Coral reef communities may actually be more likely to thrive in waters that are relatively low in nutrients.
- b.) The nutrients on which coral reef communities thrive are only found in shallow waters.
- c.) Human population growth has led to changing ocean temperatures, which threatens coral reef communities.
- d.) The growth of coral reef communities tends to destabilize underwater herbivore populations.

101. The author refers to “filter-feeding animals” in order to _____.

- a.) provide an example of a characteristic sign of reef deterioration
- b.) explain how reef communities acquire sustenance for survival
- c.) identify a factor that helps herbivore populations thrive
- d.) indicate a cause of decreasing nutrient input in waters that reefs inhabit

102. According to the passage, which of the following is a factor that is threatening the survival of coral reef communities?

- a.) The waters they inhabit contain few nutrient resources
- b.) A decline in nutrient input is disrupting their symbiotic relationship with zooxanthellae.
- c.) The degraded waters of their marine habitats have reduced their ability to carry out photosynthesis.
- d.) Waste by-products result in an increase in nutrient input to reef communities.

103. It can be inferred from the passage that the author describes coral reef communities as paradoxical most likely for which of the following reasons?

- a.) They are thriving even though human activities have depleted the nutrients in their environment.
- b.) They are able to survive in spite of an overabundance of algae inhabiting their waters.
- c.) They are able to survive in an environment with limited food resources.
- d.) Their metabolic wastes contribute to the degradation of the waters that they inhabit.

The unpleasant feeling passed and she glanced up at him. He was walking unmarked in moonlight, innocent of her reaction to him. She felt then - this thought had come to her before - that there might be more to him than she had imagined. She felt ashamed she never thanked him for the help he had given her father.

104. She glanced at him when _____.

- a.) He walked along and unnoticed in moonlight
- b.) She was sure that she was not being noticed
- c.) Her reactions did not have any effect on him
- d.) The unpleasant feeling passed

105. Her unpleasant feeling passed when _____.

- a.) He did not take any notice of her
- b.) The moonlight was beautiful
- c.) She realized her sense of shame
- d.) She looked carefully at him

106. She was ashamed because _____.

- a.) She was spying on him
- b.) There was more to him than she had imagined
- c.) A recurring thought came back to her
- d.) She had never thanked him for this help to her father

The man sat up in the snow for a moment and struggled for calmness. Then he pulled on his gloves by means of these teeth, and got upon his feet. He glanced down at first in order to assure himself that he was really standing up, for the absence of sensation in his feet left him unrelated to the earth. His erect position in itself started to drive the webs of suspicion from the dog's mind; and when he spoke peremptorily, with the sound of whip-lashes in his voice, the dog rendered its customary allegiance and came to him. As it came within reaching distance, the man lost his control. His arms flashed out to the dog and he experienced genuine surprise when he discovered that his hands could not clutch, that there was neither bend nor feeling in the fingers. He had forgotten for the moment that they were frozen and that they were freezing more and more. All this happened quickly and before the animal could get away, he encircled its body with his arms. He sat down in the snow and in this fashion held the dog, while it snarled and whined and struggled.

107. From the passage, which group of words expresses the effect of snow upon the man's feet
- a.) With the sound of whip-lashed in his voice
 - b.) His arms, flashed out to the dog
 - c.) The absence of sensation in his feet left him unrelated to the earth
 - d.) The man sat up in the snow for a moment and struggled for calmness
108. The statement that the man experienced genuine surprise when he discovered that his hands could not clutch means that _____.
- a.) The man did not see anything to clutch
 - b.) The man had nothing to clutch
 - c.) The man was afraid of the dog
 - d.) There was neither bend nor feeling in the fingers
109. Which word or group of words shows the extract condition of being 'frozen'?
- a.) Whip-lashes in his voice

- b.) He pulled on his gloves
- c.) His hands could not clutch
- d.) Lost his control

Socrates used to move about the roads or stand in the market place all day long. He talked to anyone who cared to greet him. He argued and asked many questions. Sometimes he left his listeners in a very confused state of mind because he seemed to be questioning, doubting or trying to change things about which people had never really thought, but had taken for granted. Socrates believed that everyone should learn to think for himself. He believed that everyone should have the power to see what was right, just, true and beautiful. He wanted Athens to be a perfect state and he believed that this could happen if every citizen educated his own mind to see what was right and noble.

110. People were confused by Socrates' questions because _____.

- a.) he asked them in the market place
- b.) the questions were addressed to anyone and everyone
- c.) he was argumentative and asked too many questions
- d.) the questions were on topics that people had never really thought about

111. Socrates believed that everyone should learn _____.

- a.) to read and write
- b.) to be wise
- c.) to think for himself
- d.) whatever was good and noble

112. Socrates thought that Athens would become a perfect state if _____.

- a.) its people attained perfection
- b.) everyone became educated in the arts
- c.) its people did great things
- d.) its people developed a sense of right and just

Young seekers after peace know that only equal trust shown to all the peoples of the earth and not just to a few of them, can lead to the healing of the wounds that tear them apart and so it is essential never to humiliate the members of a nation whose leaders have committed inhuman acts. Essential also is boundless concern for so many men and women who today, as exiles or

immigrants, live on foreign soil. If every home was open to somebody of foreign origin, the racial problem would be partially solved.

113. For the reconciliation and unity, it is essential _____.
- a.) to have no discrimination in the trust shown to peoples of the earth
 - b.) to have young seekers after peace
 - c.) to have confidence in at least a few of the peoples
 - d.) to punish the leaders who committed inhuman sins
114. To make the members of a nation responsible for the inhuman acts of their leaders is _____.
- a.) unimportant
 - b.) understandable
 - c.) unjust
 - d.) undesirable
115. Boundless concern should be shown to _____.
- a.) one's own countrymen alone
 - b.) the people belonging to our friendly nations only
 - c.) the foreigners who just visit our country
 - d.) all the exiles living away from their native lands
116. The theme of the passage is _____.
- a.) Rigid nationalism alone can help peaceful co-existence
 - b.) Trusting all the peoples of the earth may lead to serious problems
 - c.) Being cautious of others is a must to live in peace
 - d.) Concern for everyone irrespective of the race or country

The Bermuda Triangle occupies a disturbing and almost unbelievable place in the world's catalog of unexplained mysteries. More than a hundred planes and ships have vanished in this area into the air since 1945, and more than a thousand lives have been lost, without a single body or even a piece wreckage from the vanishing planes or ships having been found. Many of the planes concerned have vanished while in normal radio contact with their base until the very moment of their disappearance, while others have radioed the most extraordinary messages, implying that they could not get their instruments to function, that their compasses were spinning, that the sky had turned yellow

and hazy on a clear day, and the ocean, which was calm nearby, didn't look right without further clarification of what was wrong.

117. One can infer from the reading that _____.

- a.) the wreckages of some ships and planes have been found in the Bermuda Triangle
- b.) the number of incidents involving lost ships is no larger than that of any other heavily traveled region of the world
- c.) the ships. and the planes couldn't contact with their base due to the lack of equipment
- d.) the first mention of disappearances in the area was made in 1945

118. It is pointed out in the passage that _____.

- a.) thousands of people lost their lives in the Bermuda Triangle in 1945
- b.) all of the disappearances happened during the day time
- c.) the Bermuda Triangle mystery was solved in 1945
- d.) the ocean floor near Bermuda, highly unexplored, is host to many strange phenomena

119. The reading mainly deals with _____.

- a.) why so many ships and planes disappear in the Bermuda Triangle
- b.) the mysterious disappearances of ships and planes in the Bermuda Triangle
- c.) the location of the Bermuda Triangle
- d.) the frequency of the disappearances in the Bermuda Triangle

Species interdependence in nature confers many benefits on the species involved, but it can also become a point of weakness when one species involved in the relationship is affected by a catastrophe. Thus, flowering plant species dependent on insect pollination, as opposed to self-pollination or wind pollination, could be endangered when the population of insect-pollinators is depleted by the use of pesticides.

In the forests of New Brunswick, for example, various pesticides have been sprayed in the past 25 years in efforts to control the spruce budworm, an economically significant pest. Scientists have now investigated the effects of the spraying of Matacil, one of the anti-budworm agents that is least toxic to insect-pollinators. They studied Matacil's effects on insect mortality in a wide variety of wild insect species and on plant fecundity, expressed as the

percentage of the total flowers on an individual plant that actually developed fruit and bore seeds. They found that the most pronounced mortality after the spraying of Matacil occurred among the smaller bees and one family of flies, insects that were all important pollinators of numerous species of plants growing beneath the tree canopy of forests.

The fecundity of plants in one common indigenous species, the red-osier dogwood, was significantly reduced in the sprayed areas as compared to that of plants in control plots where Matacil was not sprayed. This species is highly dependent on the insect-pollinators most vulnerable to Matacil. The creeping dogwood, a species similar to the red-osier dogwood, but which is pollinated by large bees, such as bumblebees, showed no significant decline in fecundity. Since large bees are not affected by the spraying of Matacil, these results add weight to the argument that spraying where the pollinators are sensitive to the pesticide used decreases plant fecundity.

The question of whether the decrease in plant fecundity caused by the spraying of pesticides actually causes a decline in the overall population of flowering plant species still remains unanswered. Plant species dependent solely on seeds for survival or dispersal are obviously more vulnerable to any decrease in plant fecundity that occurs, whatever its cause. If, on the other hand, vegetative growth and dispersal (by means of shoots or runners) are available as alternative reproductive strategies for a species, then decreases in plant fecundity may be of little consequence. The fecundity effects described here are likely to have the most profound impact on plant species with all four of the following characteristics: a short life span, a narrow geographic range, an incapacity for vegetative propagation, and a dependence on a small number of insect-pollinator species. Perhaps we should give special attention to the conservation of such plant species since they lack key factors in their defenses against the environmental disruption caused by pesticide use.

120. Which of the following best summarizes the main point of the passage?
- a.) Species interdependence is a point of weakness for some plants, but is generally beneficial to insects involved in pollination
 - b.) Efforts to control the spruce budworm have had deleterious effects on the red-osier dogwood
 - c.) The use of pesticides may be endangering certain plant species dependent on insects for pollination
 - d.) Plant species lacking key factors in their defenses against human environmental disruption will probably become extinct

121. According to the author, a flowering plant species whose fecundity has declined due to pesticide spraying may not experience an overall population decline if the plant species can do which of the following?
- a.) Reproduce itself by means of shoots and runners
 - b.) Survive to the end of the growing season
 - c.) Survive in harsh climates
 - d.) Respond to the fecundity decline by producing more flowers
122. The passage suggests that the lack of an observed decline in the fecundity of the creeping dogwood strengthens the researcher's conclusions regarding pesticide use because the _____.
- a.) creeping dogwood is a species that does not resemble other forest plants
 - b.) creeping dogwood is a species pollinated by a broader range of insect species than are most dogwood species
 - c.) creeping dogwood grows primarily in regions that were not sprayed with pesticide, and so served as a control for the experiment
 - d.) creeping dogwood is similar to the red-osier dogwood, but its insect pollinators are known to be insensitive to the pesticide used in the study
123. The passage suggests that which of the following is true of the forest regions in New Brunswick sprayed with most anti-budworm pesticides other than Matacil?
- a.) The fecundity of some flowering plants in those regions may have decreased to an even greater degree than in the regions where Matacil is used
 - b.) Insect mortality in those regions occurs mostly among the larger species of insects, such as bumblebees
 - c.) The number of seeds produced by common plant species in those regions is probably comparable to the number produced where Matacil is sprayed
 - d.) Many more plant species have become extinct in those regions than in the regions where Matacil is used.
124. It can be inferred that which of the following is true of plant fecundity as it is defined in the passage?
- a.) A plant's fecundity decreases as the percentage of unpollinated flowers on the plant increases
 - b.) A plant's fecundity decreases as the number of flowers produced by the plant decreases

- c.) A plant's fecundity increases as the number of flowers produced by the plant increases
- d.) A plant's fecundity is usually low if the plant relies on a small number of insect species for pollination.

125. It can be inferred from the passage that which of the following plant species would be LEAST likely to experience a decrease in fecundity as a result of the spraying of a pesticide not directly toxic to plants?

- a.) A flowering tree pollinated by only a few insect species
- b.) A kind of insect-pollinated vine producing few flowers
- c.) A wind-pollinated flowering tree that is short-lived
- d.) A flowering shrub pollinated by a large number of insect species

126. Which of the following assumptions most probably underlies the author's tentative recommendation?

- a.) Human activities that result in environmental disruption should be abandoned
- b.) The use of pesticides is likely to continue into the future
- c.) It is economically beneficial to preserve endangered plant species
- d.) Conservation efforts aimed at preserving a few well-chosen species are more cost-effective than are broader-based efforts to improve the environment

By the time a child is six or seven she has all the essential avoidances well enough by heart to be trusted with the care of a younger child. And she also develops a number of simple techniques. She learns to weave firm square balls from palm leaves, to make pinwheels of palm leaves or frangipani blossoms, to climb a coconut tree by walking up the trunk on flexible little feet, to break open a coconut with one firm well-directed blow of a knife as long as she is tall, to play a number of group games and sing the songs which go with them, to tidy the house by picking up the litter on the stony floor, to bring water from the sea, to spread out the copra to dry and to help gather it in when rain threatens, to go to a neighboring house and bring back a lighted faggot for the chief's pipe or the cook-house fire.

But in the case of the little girls all these tasks are merely supplementary to the main business of baby-tending. Very small boys also have some care of the younger children, but at eight or nine years of age they are usually relieved of it. Whatever rough edges have not been smoothed off by this responsibility for younger children are worn off by their contact with older boys.

For little boys are admitted to interesting and important activities only so long as their behavior is circumspect and helpful. Where small girls are brusquely pushed aside, small boys will be patiently tolerated and they become adept at making themselves useful. The four or five little boys who all wish to assist at the important, business of helping a grown youth lasso reef eels, organize themselves into a highly efficient working team; one boy holds the bait, another holds an extra lasso, others poke eagerly about in holes in the reef looking for prey, while still another tucks the captured eels into his lavalava.

The small girls, burdened with heavy babies or the care of little staggerers who are too small to adventure on the reef, discouraged by the hostility of the small boys and the scorn of the older ones, have little opportunity for learning the more adventurous forms of work and play. So while the little boys first undergo the chastening effects of baby-tending and then have many opportunities to learn effective cooperation under the supervision of older boys, the girls' education is less comprehensive. They have a high standard of individual responsibility, but the community provides them with no lessons in cooperation with one another. This is particularly apparent in the activities of young people: the boys organize quickly; the girls waste hours in bickering, innocent of any technique for quick and efficient cooperation.

127. The primary purpose of the passage with reference to the society under discussion is to _____.
- a.) explain some differences in the upbringing of girls and boys
 - b.) criticize the deficiencies in the education of girls
 - c.) give a comprehensive account of a day in the life of an average young girl
 - d.) delineate the role of young girls
128. The word 'brusquely' most nearly means _____.
- a.) quickly
 - b.) gently
 - c.) nonchalantly
 - d.) abruptly
129. The list of techniques in paragraph one could best be described as _____
- a.) household duties
 - b.) rudimentary physical skills
 - c.) important responsibilities
 - d.) useful social skills
130. It can be inferred that the 'high standard of individual responsibility' is _____

- a.) developed mainly through child-care duties
- b.) only present in girls
- c.) taught to the girl before she is entrusted with babies
- d.) actually counterproductive

131. The expression 'innocent of' is best taken to mean _____.

- a.) not guilty of
- b.) unskilled in
- c.) unsuited for
- d.) uninvolved in

132. It can be inferred that in the community under discussion all of the following are important except _____.

- a.) domestic handicrafts
- b.) well-defined social structure
- c.) fishing skills
- d.) formal education

133. Which of the following if true would weaken the author's contention about 'lessons in cooperation'?

- I. Group games played by younger girls involve cooperation
- II. Girls can learn from watching boys cooperating
- III. Individual girls cooperate with their mothers in looking after babies

- a.) I only
- b.) II only
- c.) III only
- d.) I and II only

134. Which of the following is the best description of the author's technique in handling her material?

- a.) Both description and interpretation of observations
- b.) Presentation of facts without comment
- c.) Description of evidence to support a theory
- d.) Generalization from a particular viewpoint

Although there is a great deal of variation within each gender, on the average men and women discuss a surprisingly different range of topics. According to some studies, women and men ranging in age from seventeen to eighty described the range of topics each discussed with friends of the same sex. Certain topics were common to both men and women: work, movies, and television proved to be frequent topics for both groups. The differences between men and women were more striking than the similarities. Female friends spent much more time discussing personal and domestic subjects, relationship problems, family, health and reproductive matters, weight, food

and clothing. Men, on the other hand, were more likely to discuss music, current events, sports and business. Women were more likely to gossip about close friends and family. By contrast, men spent more time gossiping about sports figures and media personalities. These differences can lead to frustration when men and women try to converse with one another.

135. It is stated in the passage that women _____.

- a.) are unwilling to discuss personal subjects
- b.) are more interested in discussing relationship problems than are men
- c.) never talk about other men and women
- d.) don't like gossiping about anything

136. According to the passage, men _____.

- a.) need to learn to communicate better
- b.) like talking about movies and television as much as women do
- c.) are not likely to gossip on anything
- d.) have no common topics with women

137. The passage mainly discusses _____.

- a.) what women's conversational topics are
- b.) why men don't like conversing with women
- c.) the topics men like discussing
- d.) the conversational topics of men and women

Most of you probably did not see Mohan at close quarters. He had amazing qualities. One of these qualities was that he managed to draw out the good in another person. The other person may have had plenty of evil in him. But he somehow spotted the good and laid emphasis on the good. The result was that the poor man had to try to be good. He could not help it. He would feel ashamed when he did something wrong.

138. The author assumes that most of us _____.

- a.) have not seen Mohan
- b.) have not heard of Mohan
- c.) are not well-acquainted with Mohan's powers

d.) have not observed Mohan's house closely

139. One of Mohan's greatest qualities was that _____.

- a.) he could completely destroy the evil in another man
- b.) he could discover the good in another man
- c.) he always maintained a good life
- d.) he always kept away from bad men

140. The other man is called poor because _____.

- a.) he usually came from a poor family
- b.) he was always very evil
- c.) he must try to become good and give up his evil ways
- d.) he did not know what to do and was helpless

141. The other man tried to be good because _____.

- a.) Mohan forced him to do so
- b.) he wanted to imitate Mohan
- c.) he was ashamed of doing a wrong thing
- d.) he did not want to remain poor

People seldom feel neutral about poetry. Those who love it sometimes give the impression that it is an adequate substitute for food, shelter, and love. But it isn't. Those who dislike poetry on principle sometimes claim, on the other hand, that poetry is only words and good for nothing. That's not true either. When words represent and recreate genuine human feelings, as they often do in poetry, they can be very important. Poems provide, in fact, a language for feeling, and one of poetry's most insistent merits involves its attempt to express the inexpressible. One of the joys of experiencing poetry occurs when we read a poem and want to say, "yes, that is just what it is like; I know exactly what that line means but I have never been able to express it so well." Poetry can be the voice of our feelings even when our minds are speechless with grief or joy.

142. One can understand from the passage that people _____.

- a.) seldom feel that poetry is an equivalent for life itself
- b.) rarely take a biased opinion about poetry
- c.) generally think that poetry expresses what might otherwise seem unutterable
- d.) generally think of poetry as extremely important or totally useless

143. One point made by the author in the passage is that poetry _____.

- a.) tends to make the reader disappointed
- b.) is an adequate substitute for food, shelter, and love
- c.) is only words and good for nothing
- d.) often captures real human feelings

144. The author points out in the reading that _____.

- a.) poetry is not closely concerned with feelings
- b.) poems are primarily about how people think rather than how people feel
- c.) poetry can't be the expression of one's deepest feelings
- d.) few people think that poetry is neutral

Albert Einstein was born in 1879 in the German city of Ulm. He had been no infant prodigy; indeed, he was so late in learning to speak that his parents feared he was fool. In school, though his teachers saw no special talent in him, the signs were already there. He taught himself calculus, for example, and he told me that his teachers seemed a little afraid of him because he asked questions they could not answer. At the age of 16, he asked himself whether a light wave would seem stationary if one ran side by side of it. It seems an innocent question, but this shows Einstein going to the heart of a problem. From it there would arise, ten years later, his theory of relativity.

145. Albert Einstein _____.

- a.) is the most genius scientist of all
- b.) was born in the 18th century
- c.) was spoiled by his teachers in school
- d.) made his parents afraid due to his problem in speech in his childhood

146. When he was in school, Einstein _____.

- a.) showed no signals about his genius
- b.) was afraid of his teachers due to their innocent questions
- c.) asked difficult questions to his teachers
- d.) was understood as genius by public when he was a teenager

147. It's obvious in the passage, Einstein's _____.

- a.) genius was a gift
- b.) character was perfect to most of his teachers
- c.) life encouraged most of the scientist in the world
- d.) question is the origin of his theory of relativity

The Richter Magnitude Scale is a method of assigning a single number to quantify the energy released during an earthquake. The magnitude is defined as the logarithm of the ratio of wave's amplitude, measured through a seismograph. Therefore, an earthquake that measures 5.0 of the Richter scale has a shaking amplitude of 10 times that of an earthquake measuring 4.0. Since the middle of the 20th century, the Richter Scale has been mostly superseded by the Moment Magnitude Scale, the Richter Scale is still used in Russia and some other former Soviet Union states.

148. Which one of the following best replaces "method" while maintaining the meaning of the sentence?
- a.) Measurement b.) Scale c. Approach d.) Process
149. Which one of the following best replaces "superseded" while maintaining the meaning of the sentence?
- a.) Supplanted b.) Transferred c.) Removed d.) Usurped

In 1892, the Sierra Club was formed. In 1908, an area of coastal redwood trees north of San Francisco was established as Muir Woods National Monument. In the Sierra Nevada Mountains, a walking trail from Yosemite Valley to Mount Whitney was dedicated in 1938. It is called the John Muir Trail.

John Muir was born in 1838 in Scotland. His family name means "moor," which is a meadow full of flowers and animals. John loved nature from the time he was small. He also liked to climb rocky cliffs and walls.

When John was 11 years old, his family moved to the United States and settled in Wisconsin. John was good with tools and soon became an inventor. He first invented a model of a sawmill. Later, he invented an alarm clock that would cause the sleeping person to be tipped out of bed when the timer sounded.

Muir left home at an early age. He took a 1,000-mile walk south to the Gulf of Mexico in 1867 and 1868. Then he sailed for San Francisco. The city was too noisy and crowded for Muir, so he headed inland for the Sierra Nevadas.

When Muir discovered the Yosemite Valley in the Sierra Nevadas, it was as if he had come home. He loved the mountains, the wildlife, and the trees. He climbed the mountains and even climbed trees during thunderstorms in order to get closer to the wind. He put forth the theory in the late 1860s that the Yosemite Valley had been formed through the action of glaciers. People ridiculed him. Not until 1930 was Muir's theory proven correct.

Muir began to write articles about the Yosemite Valley to tell readers about its beauty. His writing also warned people that Yosemite was in danger from timber mining and sheep ranching interests. In 1901, Theodore Roosevelt became president of the United States. He was interested in conservation. Muir took the president through Yosemite, and Roosevelt helped get legislation passed to create Yosemite National Park in 1906.

Although Muir won many conservation battles, he lost a major one. He fought to save the Hetch Hetchy Valley, which people wanted to dam in order to provide water for San Francisco. In late 1913, a bill was signed to dam the valley. Muir died in 1914. Some people say losing the fight to protect the valley killed Muir.

150. What happened first?

- a.) The Muir family moved to the United States
- b.) Muir Woods was created
- c.) John Muir learned to climb rocky cliffs
- d.) John Muir walked to the Gulf of Mexico

151. When did Muir invent a unique form of alarm clock?

- a.) While the family still lived in Scotland
- b.) After he sailed to San Francisco
- c.) After he traveled in Yosemite
- d.) While the Muir family lived in Wisconsin

152. What did John Muir do soon after he arrived in San Francisco?

- a.) He ran outside during an earthquake
- b.) He put forth a theory about how Yosemite was formed
- c.) He headed inland for the Sierra Nevadas
- d.) He began to write articles about the Sierra Nevadas

153. When did John Muir meet Theodore Roosevelt?

- a.) Between 1901 and 1906
- b.) Between 1838 and 1868
- c.) Between 1906 and 1914
- d.) Between 1868 and 1901

154. What happened last?

- a.) John Muir died
- b.) John Muir Trail was dedicated
- c.) Muir's glacial theory was proven
- d.) The Sierra Club was formed

Life processes are similar in all animals with backbones. So physiologists can find out a lot about the human body by studying animals such as frogs, rats and rabbits. Sometimes research involves the death of these animals. But physiologists take great care not to hurt the animals. The animals are usually put to sleep during the experiments so they do not feel pain. Many people think animals should not be used for experiments. But it is only by physiological research that medicine has been able to advance. Polio is a disease that attacks the nerves. Millions of people have been protected from polio by a fluid called a vaccine. Tissue from monkeys is used to make the vaccine.

155. According to the passage,

- a.) animals with bones are of no great importance in experimental researches
- b.) animals can die if not vaccinated by experimental team
- c.) although many people resist against animals being utilized in researches, it is necessary
- d.) by the help of experimental researches, millions of people have been protected against rats and frogs

156. Experimental researches _____.

- a.) are of paramount significance for the polio infected monkeys
- b.) are the mere way that medicine can progress
- c.) are carried out to remove infected tissues from the animals

d.) include only reptiles to discover the vaccines

157. It's clearly stated in the passage that

- a.) millions of people have been killed by polio disease so far
- b.) physiologists can cure rats, rabbits and frogs as well as people
- c.) experimental animals are first put to sleep and then used for research
- d.) monkeys are usually infected with polio

In the 16th century, an age of great marine and terrestrial exploration, Ferdinand Magellan led the first expedition to sail around the world. As a young Portuguese noble, he served the king of Portugal, but he became involved in the quagmire of political intrigue at court and lost the king's favor. After he was dismissed from service by the king of Portugal, he offered to serve the future Emperor Charles V of Spain.

A papal decree of 1493 had assigned all land in the New World west of 50 degrees W longitude to Spain and all the land east of that line to Portugal. Magellan offered to prove that the East Indies fell under Spanish authority. On September 20, 1519, Magellan set sail from Spain with five ships. More than a year later, one of these ships was exploring the topography of South America in search of a water route across the continent. This ship sank, but the remaining four ships searched along the southern peninsula of South America. Finally, they found the passage they sought near 50 degrees S latitude. Magellan named this passage the Strait of All Saints, but today it is known as the Strait of Magellan.

One ship deserted while in this passage and returned to Spain, so fewer sailors were privileged to gaze at that first panorama of the Pacific Ocean. Those who remained crossed the meridian now known as the International Date Line in the early spring of 1521 after 98 days on the Pacific Ocean. During those long days at sea, many of Magellan's men died of starvation and disease.

Later, Magellan became involved in an insular conflict in the Philippines and was killed in a tribal battle. Only one ship and 17 sailors under the command of the Basque navigator Elcano survived to complete the westward journey to Spain and thus prove once and for all that the world is round, with no precipice at the edge.

158. The 16th century was an age of great _____ exploration.

- a.) cosmic b.) land c.) mental d.) common man

159. Magellan lost the favor of the king of Portugal when he became involved in a political _____.

- a.) entanglement b.) discussion c.) negotiation d.) problem

160. The Pope divided New World lands between Spain and Portugal according to their location on one side or the other of an imaginary geographical line 50 degrees west of Greenwich that extends in a _____ direction.

- a.) north and south b.) crosswise c.) easterly d.) southeast

161. One of Magellan's ships explored the _____ of South America for a passage across the continent.

- a.) coastline b.) mountain range c.) physical features d.) islands

162. Four of the ships sought a passage along a southern _____.

- a.) coast c.) body of land with water on three sides
b.) inland d.) border

163. The passage was found near 50 degrees S of _____.

- a.) Greenwich b.) The equator c.) Spain d.) Portugal

164. In the spring of 1521, the ships crossed the _____ now called the International Date Line.

- a.) imaginary circle passing through the poles
b.) imaginary line parallel to the equator
c.) area
d.) land mass

For nearly twenty years I taught boys, lovingly and was being loved in return. When, after twenty-eight years of political work, I returned to education, I might have confined myself to administrative side, but took part in the actual instruction. This I did because I found happiness in it. Modern

conditions do not tolerate caste and its monopolies, and the high calling of the educator is open to all.

165. The author suggests that he left teaching after twenty years because _____
- a.) he was dissatisfied with teaching
 - b.) he wanted to become administrator
 - c.) he wanted to take part in political activities
 - d.) he was busy with family and domestic affairs
166. Which of the following types of educational activities did the author take up?
- a.) Educational administration
 - b.) Teaching
 - c.) Advising educational institutions
 - d.) All the above
167. Which one of the following phrases best helps to bring out the precise meaning of the term 'high calling of the educator'?
- a.) The noble profession of teaching
 - b.) The serious advice to the teacher
 - c.) The difficult work of educational administration
 - d.) The high-sounding phrases used to describe a teacher

He was that rare figure: a natural-born leader, tailored for greatness. When he was but a child his mother had whispered to him, "My son, you are a child of the dawn. You will be a man of glory, a great leader of your people." Sukarno never forgot those words. Liberating his land from 350 years of colonial rule, he molded many disparate peoples into a nation that became the fifth most populous in the world, one of the richest of all in natural resources. He gave his people a language of their own, raised their literacy rate from six to above 55 percent, gave them an intense pride in being Indonesians. He had monumental talents: an oratorical gift that could sway and charm, an abundance of that indefinable mystique called "charisma." Responding, his people gave him loyalty beyond love, reverence approaching god-worship.

168. Sukarno is a man who _____.
- a.) is endowed with leadership
 - b.) used to design clothes when he was a child

- c.) had a natural-born great mother
- d.) caused people to be rebellious against the government

169. It's stated in the passage that Sukarno _____.

- a.) is the only child of his parents
- b.) couldn't liberate his land from colonial rule
- c.) built a nation out of separate communities
- d.) opposed giving his public their own language

170. After having freed his land from colonial rule, Sukarno _____.

- a.) suddenly built a monument dedicated to himself
- b.) caused his people to be ashamed of themselves
- c.) increased the population of Indonesia
- d.) increased the level of literacy of his people

Although genetic mutations in bacteria and viruses can lead to epidemics, some epidemics are caused by bacteria and viruses that have undergone no significant genetic change. In analyzing the latter, scientists have discovered the importance of social and ecological factors to epidemics. Poliomyelitis, for example, emerged as an epidemic in the United States in the twentieth century; by then, modern sanitation was able to delay exposure to polio until adolescence or adulthood, at which time polio infection produced paralysis.

Previously, infection had occurred during infancy, when it typically provided lifelong immunity without paralysis. Thus, the hygiene that helped prevent typhoid epidemics indirectly fostered a paralytic polio epidemic. Another example is Lyme disease, which is caused by bacteria that are transmitted by deer ticks. It occurred only sporadically during the late nineteenth century but has recently become prevalent in parts of the United States, largely due to an increase in the deer population that occurred simultaneously with the growth of the suburbs and increased outdoor recreational activities in the deer's habitat. Similarly, an outbreak of dengue hemorrhagic fever became an epidemic in Asia in the 1950's because of ecological changes that caused *Aedes aegypti*, the mosquito that transmits the dengue virus, to proliferate. The stage is now set in the United States for a dengue epidemic because of the inadvertent introduction and wide dissemination of another mosquito, *Aedes albopictus*.

171. The passage suggests that a lack of modern sanitation would make which of the following most likely to occur?
- a.) An outbreak of Lyme disease
 - b.) An outbreak of dengue hemorrhagic fever
 - c.) An epidemic of typhoid
 - d.) An epidemic of paralytic polio among infants
172. According to the passage, the outbreak of dengue hemorrhagic fever in the 1950's occurred for which of the following reasons?
- a.) The mosquito *Aedes aegypti* was newly introduced into Asia
 - b.) The mosquito *Aedes aegypti* became more numerous
 - c.) The mosquito *Aedes albopictus* became infected with the dengue virus
 - d.) Individuals who would normally acquire immunity to the dengue virus as infants were not infected until later in life
173. It can be inferred from the passage that Lyme disease has become prevalent in parts of the United States because of which of the following?
- a.) The inadvertent introduction of Lyme disease bacteria to the United States
 - b.) The inability of modern sanitation methods to eradicate Lyme disease bacteria
 - c.) A genetic mutation in Lyme disease bacteria that makes them more virulent
 - d.) An increase in the number of humans who encounter deer ticks
174. Which of the following can most reasonably be concluded about the mosquito *Aedes albopictus* on the basis of information given in the passage?
- a.) It is native to the United States
 - b.) It can proliferate only in Asia
 - c.) It transmits the dengue virus
 - d.) It caused an epidemic of dengue hemorrhagic fever in the 1950's.
175. Which of the following best describes the organization of the passage?
- a.) A paradox is stated, discussed and left unresolved
 - b.) Two opposing explanations are presented, argued, and reconciled
 - c.) A theory is proposed and is then followed by descriptions of three experiments that support the theory
 - d.) A generalization is stated and is then followed by three instances that support the generalization

SCIENCE

176. "For every action, there is an equal and opposite reaction." If a baseball player hits a ball with a bat – and the "action" is said to be the impact of the bat against the ball – what is its "opposite reaction?"
- The grip of the player's hand on the bat
 - The air resistance on the ball
 - The force of the ball against the bat
 - Muscular effort in the player's arm
177. Gravitational potential energy is energy an object possesses because of its position in a gravitational field. Which of the following has the most gravitational potential energy?
- A truck at the top of the hill
 - A truck speeding down the hill
 - A pendulum at the center of trajectory
 - A man on his mountain bike speeding down the hill
178. Atoms of an element differ from those of all elements in _____.
- atomic number and electronic configuration
 - number of neutrons and number of valence electrons
 - atomic number and number of valence electrons
 - number of neutrons and electronic configuration
179. In the chemical equation $N_2 + 3H_2 \rightarrow 2NH_3$, how many moles of H_2 are needed to react with 2.5 mole N_2 ?
- 2.5
 - 5.0
 - 7.5
 - 10.0
180. Balance the following equation: $C_6H_{12}O_4 + O_2 \rightarrow CO_2 + H_2O$
- $C_6H_{12}O_4 + 15O_2 \rightarrow 6CO_2 + 7H_2O$
 - $C_6H_{12}O_4 + 7O_2 \rightarrow 6CO_2 + 6H_2O$
 - $4C_6H_{12}O_4 + 30O_2 \rightarrow 24CO_2 + 26H_2O$
 - $C_6H_{12}O_4 + 15O_2 \rightarrow 6CO_2 + 7H_2O$
181. A new insecticide was very effective in killing flies and mosquitoes. After several months, fewer insects were observed to die from the spray. The reason why fewer insects are killed is that _____.
- Insects which survive the spraying developed a liking for the insecticide
 - Insects which survive the spraying developed immunity to the insecticide

- c.) The insecticide's effectiveness was reduced with aging
- d.) There is permanent mutation giving insects resistance to the insecticide

182. Which of the following is true about sub atomic particles, mass number and atomic number?

- a.) mass number is equal to the number of neutrons
- b.) proton plus electron is equal to the mass number
- c.) atomic number is equal to the number of protons
- d.) neutron number can be calculated given only the mass number

183. Why is a virus not a living organism?

- a.) It can never reproduce
- b.) It is simpler than bacteria
- c.) It is not a complete cell
- d.) All of the above

184. When mechanical energy is conserved, potential energy is maximum when kinetic energy is

- a.) maximum
- b.) minimum
- c.) equal
- d.) none of the above

185. In an atomic explosion, enormous energy is released which is due to _____

- a.) conversion of chemical energy into heat energy
- b.) conversion of mechanical energy into nuclear energy
- c.) conversion of mass into energy
- d.) conversion of neutrons into protons

186. What differentiates a prokaryote from a eukaryote?

- a.) a eukaryotic cell possesses a genetic material while a prokaryotic cell does not.
- b.) both have a membrane system responsible for delineating the cell from its surroundings
- c.) a eukaryotic cell consists of a true nucleus that is membrane-bound while a prokaryotic cell does not
- d.) algae and yeasts are examples of organisms made up of eukaryotic cells while bacteria are prokaryotic in nature

187. If the pressure exerted on a confined ideal gas is doubled, what happens to its temperature and volume, taken separately?

- a.) doubled; halved
- b.) remains the same

- c.) halved; halved
- d.) halved; doubled

188. What is the kinetic energy of a 500-kg car moving at 3.0 m/s?

- a.) 1.50 kJ
- b.) 1.75 kJ
- c.) 2.00 kJ
- d.) 2.25 kJ

189. Given two vectors $A = 4m\hat{i} + 3m\hat{j}$ and $B = 2m\hat{i} - 3m\hat{j}$, what is $A + B$?

- a.) $8.61m$
- b.) $6m\hat{i}$
- c.) $6m$
- d.) $8.61m\hat{i}$

190. A stone is thrown into the air at an angle. Neglecting air resistance, what is the stone's velocity upon reaching its maximum height?

- a.) zero
- b.) equal to the horizontal component of its initial velocity
- c.) equal to the vertical component of its initial velocity
- d.) equal to the acceleration due to gravity

191. A mixture shows the following properties: Its particles do not settle down, can not be filtered, and cannot be seen by the naked eye. The mixture does not show the Tyndall effect. Which of the following best describe this mixture?

- a.) homogeneous
- b.) solution
- c.) colloid
- d.) suspension

192. Salt water has a density greater than that of pure water. What does this mean?

- a.) Salt water is heavier than water
- b.) Salt water is lighter than water
- c.) Salt water is as heavy as water
- d.) Salt water has more volume than water

193. Weathering is the process that changes solid rock into sediments. With weathering, rock is disintegrated. It breaks into pieces. However, rocks on the moon are not being weathered. Which of the following is the reason for this?

- a.) There is no atmosphere on the moon
- c.) There is no volcano on the moon

b.) There is no water on the moon

d.) The gravity on the moon is very small

194. What is probability that a father with a homogenous dominant trait and a mother with the same homogeneous dominant trait will produce an offspring with the recessive trait?

a.) 25%

b.) 15%

c.) 0%

d.) 100%

195. The velocity of a car changes from 70 m/s north to 50 m/s north in 5 seconds. The magnitude of the car's acceleration is _____.

a.) 24 m/s^2

b.) 4 m/s^2

c.) 4 m/s^2

d.) 24 m/s^2

196. When an iron nail gets rusted, iron oxide is formed _____.

a.) without any change in the weight of the nail

b.) with decrease in the weight of the nail

c.) with increase in the weight of the nail

d.) without any change in color and weight of the nail

197. What force is needed to make a 2 kg rock accelerate at 0.5 m/s^2 ?

a.) 1 N

b.) 2 N

c.) 3 N

d.) 4 N

198. What will happen if the white blood cells are deteriorating?

a.) Oxygen cells will not be transported

b.) The nerve impulses will not be carried to the brain

c.) The harmful bacteria in the body will not be destroyed

d.) There will be no production of hormone to clot the blood

199. Metal atoms tend to _____ while nonmetal atoms tend to _____.

a.) lose electrons; gain protons

c.) lose electrons; gain electrons

b.) gain protons; lose electrons

d.) gain electrons; lose protons

200. The primary difference between RNA and DNA is that _____.

a.) DNA is single-stranded and RNA is double-stranded

b.) DNA is a double-stranded helix and RNA is single-stranded

- c.) The sugar in DNA is ribose
- d.) The sugar in both RNA and DNA is deoxyribose

201. When you go to a higher altitude, what happens to the boiling point of the water?

- a.) increases b.) stays the same c.) decreases d.) fluctuates

202. Oil and water are immiscible such that when both are placed in a single container oil “floats” on top of the water. Why is this so?

- a.) Oil is a nonpolar substance while water is polar
- b.) Oil is a polar substance while water is a nonpolar substance
- c.) Oil is less dense than water
- d.) Oil is denser than water

203. Aside from the properties of the molecules involved, the principal factors that affect the speed of a chemical reaction are:

- I. presence of catalyst
- II. the temperature at which the reaction occurs
- III. the concentration of the reactants
- turbulence in the reacting medium

- a.) II, III b.) I, III c.) III d.) none of the above

204. If both the force and area of application of the said force are halved, the pressure will _____.

- a.) increase by a factor of 2 c.) diminish by square root of 2
- b.) decrease by a factor of 2 d.) remain constant

205. Which of the following does not describe a prokaryotic cell?

- a.) Their DNA is not bound by a nuclear membrane
- b.) They do not possess a true nucleus
- c.) A plasma membrane surrounds and encloses the prokaryotic cell
- d.) They possess an extensive endoplasmic reticulum

206. Which of the following equations does not belong to the group?

- a.) $C + O_2 \rightarrow CO_2$ c.) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
- b.) $2KClO_3 \rightarrow 2KCl + 3O_2$ d.) $2Mg + O_2 \rightarrow 2MgO$

207. Which of the following statement is true?
- a.) Orbitals make up a subshell; subshells make up a shell
 - b.) Subshells make up a shell; shells make up an orbital
 - c.) Shells make up an orbital; orbitals make up a subshell
 - d.) None of the above
208. How much time should it take for a traveling bullet to hit the ground compared to a bullet dropped from rest?
- a.) The traveling bullet will take longer to hit the ground than the bullet dropped from rest
 - b.) The traveling bullet will hit the ground at the same time as the bullet dropped from rest
 - c.) The traveling bullet will hit the ground before the bullet dropped from rest
 - d.) The traveling bullet moves so fast that it will never hit the ground
209. The moon is said to light up the night skies. What gives the moon this ability to shine?
- a.) The sun's reflection off the moon
 - b.) Radiation is produced by the moon's atmosphere
 - c.) Radiation from the Sun that is absorbed by the moon
 - d.) The luminous substance found on the moon's surface
210. Which of the following is not a chemical reaction?
- a.) burning paper
 - b.) rusting of metal
 - c.) ripening of fruit
 - d.) freezing carbon dioxide
211. How does the esophagus bring food down to the stomach in space, where there is zero gravity?
- a.) There is micro-gravity in space caused by the centrifugal force created by orbiting around Earth
 - b.) Peristalsis
 - c.) Mucous membranes in the esophagus help the food go down
 - d.) Food is liquefied and makes use of osmosis in the esophagus to bring down food.
212. Which of the following will NOT increase the speed of a reaction?

- a.) Catalyst b.) Heat c.) Carbonates in water d.) Increasing surface area

213. If the mass of the Earth were twice as it is now, the gravitational force between it and the sun would be _____.

- a.) twice as great b.) thrice as great c.) the same d.) four times as great

214. What happens to the amount of energy as the level rises in the food pyramid?

- a.) increases b.) decreases c.) remains constant d.) decrease then increase

215. In which population would you expect the most rapid evolutionary change?

- a.) a small population with a high mutation rate in a changing environment
b.) a small population with a low mutation rate in a stable environment
c.) a large population with a high mutation rate in a changing environment
d.) a large population with a low mutation rate in a stable environment

216. Global climate change can be attributed to the increase in what two gases produced by human activities?

- a.) nitrous oxide and sulfur dioxide c.) ozone and carbon dioxide
b.) methane and carbon dioxide d.) ozone and methane

217. In an atomic nucleus, neutrons and protons are held together by

- a.) gravitational forces c.) coulombic forces
b.) exchange forces d.) magnetic forces

218. Which of the statements below does not explain the movements of the earth's crust?

- a.) Earthquakes that originate beneath the sea can produce tsunamis
b.) The mantle beneath the earth's crust is made of plastic materials that have convection currents
c.) The earth's crust has cracked into huge plates which move slowly due to convection currents in the mantle

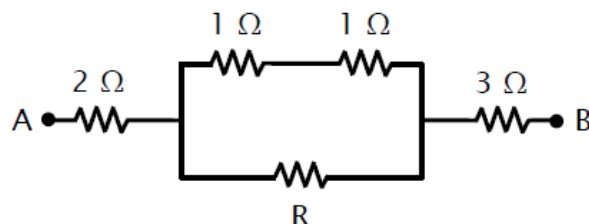
- d.) Many years of stress on the rock may fracture the crust causing one portion to slide upward or downward with respect to the other
219. A relationship between two living things by which one is benefited while the other is neither harmed nor benefited is called ____.
- a.) Commensalism b.) Parasitism c.) Mutualism d.) Competition
220. Cyanogen (C_2N_2) can be prepared by a catalyzed phase reaction between HCN and NO_2 . The products of the reaction are C_2N_2 , NO , and H_2O . What is the balanced chemical equation?
- a.) $4HCN + 2NO_2 \rightarrow 2C_2N_2 + 4NO + 2H_2O$
 b.) $2HCN + NO_2 \rightarrow C_2N_2 + NO + H_2O$
 c.) $4HCN + NO_3 \rightarrow 2C_2N_2 + NO + 2H_2O$
 d.) $2HCN + 3NO_2 \rightarrow C_2N_2 + 3NO + H_2O$
221. Simple, multicellular animals with tissues but no distinct organs. Commonly known as sponges, they typically attach to rocks, shells, or coral. What are they called?
- a.) Placozoa b.) Porifera c.) Cnidaria d.) Ctenophora
222. A baseball bat exerts a magnitude of force of 60 N on a ball. What is the magnitude of force exerted by the ball on the bat?
- a.) 60 N b.) 180 N c.) 120 N d.) 20 N
223. Which among the following does not belong to the group?
- a.) Speed b.) Temperature c.) Time d.) Velocity
224. Newton's first law of motion states that "Every object remains at rest or in motion in a straight line at constant speed unless acted upon by an unbalanced force." What is the net force acting on an airplane in level flight at 500 kph due north?
- a.) 100 N b.) 490 N c.) 980 N d.) 0 N
225. Which is not true about moon and tides?

- a.) When the moon is close to earth, the lesser the effect of gravity
- b.) When the moon and earth are in line, the highest tides occur on earth
- c.) When earth and moon are at right angles, the lowest tides occur on earth
- d.) There are two high tides at opposite sides of the earth and two low tides at the other

226. The tendency of an object to resist changes in motion is dependent on

- a.) Weight
- b.) Temperature
- c.) Speed
- d.) Mass

227. The figure below shows resistors connected in series and parallel. If the total resistance from A to B is $8\ \Omega$, find the resistance of R.



- a.) $1\ \Omega$
- b.) $2\ \Omega$
- c.) $3\ \Omega$
- d.) $4\ \Omega$

228. A rally driver has 5 seconds to stop his car which traveling at a speed of 20 m/s. What is his average acceleration?

- a.) $4\ m/s^2$
- b.) $3\ m/s^2$
- c.) $1\ m/s^2$
- d.) $0.25\ m/s^2$

229. Most plants are not suitable for human consumption. This is because we do not possess the ability to digest them properly. Why is this so?

- a.) They are poisonous
- b.) We can't chew them properly
- c.) They have a high concentration of cellulose
- d.) Plants have an anti-digestion substance in them

230. Which of the following is NOT a chemical change?

- a.) Compression of carbon
- b.) Mixing zinc and HCl
- c.) Burning of wood
- d.) Decomposition of leaves

231. A heavy ball and a plastic ball of equal diameters were dropped from a height of 30 meters. Which will hit the ground first?

- a.) They will hit the ground at the same time
- b.) The heavy ball will hit the ground first
- c.) The plastic ball will hit the ground first
- d.) It depends on the air under each ball

232. Dante walks 0.5 km north. Starting from the same point, Richie walks 1.2 km east. What would be the distance separating the two boys?

- a.) 0.7 km
- b.) 0.9 km
- c.) 1.1 km
- d.) 1.3 km

233. Gon wants to know which is stronger, Bisuke's punch or Killua's. Using the scientific method, how should Gon setup his controlled setup?

- a.) Let Bisuke and Killua fight till one of them wins
- b.) Let Bisuke and Killua punch him then compare which is more painful
- c.) Let Bisuke and Killua punch the same object a number of times then compare the damage done
- d.) None of the above

234. An advantage of sexual reproduction over asexual reproduction is that it:

- a.) increases mutation rate
- b.) reduces the offspring's risk of death during development
- c.) increases genetic variability in a population
- d.) all of the above

235. Two populations may or may not affect each other. What do you call the interaction where in neither population affect each other?

- a.) Mutualism
- b.) Competition
- c.) Commensalism
- d.) Neutralism

MATHEMATICS

236. Jeb is 14 years younger than Ed. In 6 years, Ed will be thrice as old as Jeb. How old is Jeb 4 years from now?

- a.) 15 b.) 19 c.) 5 d.) 9

237. Thrice the larger of two numbers is five more than five times the smaller, and the sum of six times the larger and five times the smaller is 265. What is the larger number?

- a.) 44 b.) 40 c.) 30 d.) 27

238. If the lengths of the sides of a square are tripled, what is the effect on its area?

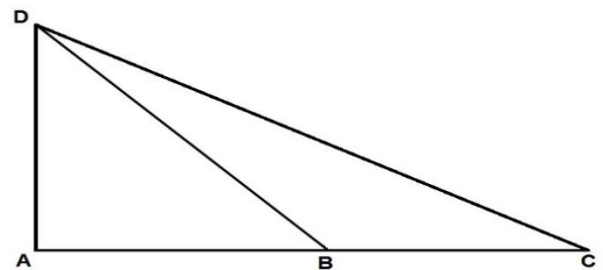
- a.) The area will increase thrice. c.) The area will increase 9 times.
b.) The area will increase 19 times. d.) The area will remain the same.

239. Simplify the following expression: $\frac{4}{\sqrt{x+3}} + \frac{\sqrt{5x}}{\sqrt{x+3}}$.

- a.) $4+5xx+3$ c.) $4+5xx+3$
b.) $4+5xx+3$ d.) $3xx+3$

240. Given the figure, $\overline{BD} = \overline{AC}$; $\overline{AD} = \overline{AB} = 8$. Find $\overline{CD}$.

- a.) $8\sqrt{2}$
b.) $8\sqrt{3}$
c.) $8 + 8\sqrt{2}$
d.) $8 + 8\sqrt{3}$

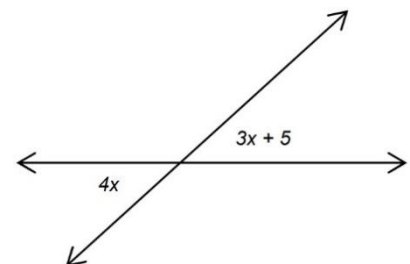


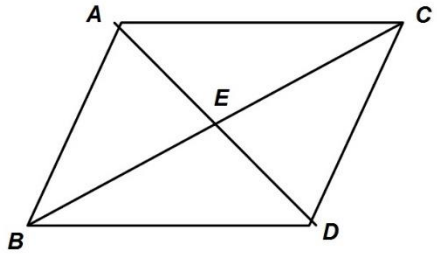
241. Divide $12x^7y^3 + 8x^6y^4 - 24x^8y^9$ by $2x^4y^2$.

- a.) $6x^3y + 4x^2y^2 - 12x^4y^7$ c.) $-6x^3y - 4x^2y^2 + 12x^4y^7$
b.) $6x^3y + 8x^2y^2 - 12x^4y^7$ d.) $-6x^3y - 8x^2y^2 + 12x^4y^7$

242. What is the value of x in the diagram shown?

- a.) 10
b.) 5
c.) $\frac{5}{7}$
d.) 7



243. What is the first term of an arithmetic sequence whose 19th and 12th terms are 417 and 319 respectively?
- a.) 165 b.) 242 c.) 98 d.) 7
244. What is the sum of the first seven terms in the arithmetic progression whose first four terms are -7, -3, 1, and 5?
- a.) 18 b.) 56 c.) 35 d.) 81
245. If x and y are positive integers where $x > y$, which of the following is always true?
- a.) $\frac{y^2}{x^2} > \frac{y}{x}$ b.) $\frac{y^2}{x^2} > \frac{x^2}{y^2}$ c.) $\frac{x^2}{y^2} > \frac{x}{y}$ d.) $\frac{x^2}{y^2} > \frac{x^3}{y^3}$
246. Suppose that $\angle ABC$ and $\angle ACB$ are equal, what is the value of x if $AE = 3x + 4$; $CE = y - 3$; $BE = 2x + 1$; $DE = 2y + 5$.
- a.) 9
b.) -9
c.) 17
d.) -17
- 
247. The sum of four consecutive multiples of 7 is 1582. What is the smallest among those integers?
- a.) 385 b.) 420 c.) 364 d.) 399
248. Kevin can finish cleaning their room for about 36 minutes. When his brother Ben helps him, it takes them 9 minutes to complete cleaning. How many minutes would it take Ben to clean their room alone?
- a.) 715 b.) 15 c.) 18 d.) 12
249. Solve for x : $\log_2(4x - 12) = 5$.
- a.) 9.25 b.) 11 c.) 3.25 d.) 13

250. A rectangular solid with dimensions 2, 8, and x has the same volume as a cube with side length of 4. What is the value of x ?
- a.) 4 b.) 2 c.) 8 d.) 16
251. Jack alone can carry a pail of water uphill for 40 minutes. Jill can do the same for 25 minutes. How long will it take for both of them go reach the peak of the hill if Jill helps his friend Jack?
- a.) 22.5 minutes b.) 15.4 minutes c.) 20 minutes d.) 17.25 minutes
252. The sum of five consecutive even integers is 470. What is the smallest among those integers?
- a.) 84 b.) 90 c.) 96 d.) 88
253. Simplify $3\sqrt{2} + 2\sqrt{18} - 5\sqrt{50}$.
- a.) $16\sqrt{2}$ b.) $-16\sqrt{2}$ c.) 0 d.) $5\sqrt{2} - 25\sqrt{3}$
254. Roy is 13 years older than Sam. In 5 years, the sum of Roy's age and twice that of Sam's age is 79. What is Sam's age now?
- a.) 30 b.) 17 c.) 32 c.) 22
255. The width of a rectangle is *thrice* the length and is equal to the side of a square. Which of these statements is *true*?
- a.) The area of the square is $1/9$ times that of the rectangle.
b.) The area of the square is $1/3$ times that of the rectangle.
c.) The area of the rectangle is $1/9$ times that of the square.
d.) The area of the rectangle is $1/3$ times that of the square.
256. Ben and Kevin arrange to meet on the highway connecting their hometowns. Ben drives at 45 kph and Kevin at 35 kph. They leave their homes which are 120 kilometers apart at the same time. In how many hours will they meet?
- a.) 1 b.) 1.5 c.) 2 d.) 2.5

257. Sam, Ton, Paul can finish a job in 3 hours. If Sam does the job alone, he can finish it in 6 hours. If Ton does the job alone, he can finish it in 8 hours. How long will it take for Paul to finish the job alone?
- a.) 12 b.) 16 c.) 24 d.) 18
258. Joy is 9 years older than Leo. Three years ago, the sum of their ages was 29. How old is Joy now?
- a.) 10 b.) 19 c.) 13 d.) 22
259. Kaye is 24 years old. Her cousin, Pia is 9 years old. In how many years will Kaye be double Pia's age?
- a.) 6 b.) 8 c.) 4 d.) 10
260. A tank contains 9000 liters of a solution that is 30% acid. How much water should be added to make a solution that is 20% acid?
- a.) 3000 liters b.) 4500 liters c.) 5000 liters d.) 5500 liters
261. Thrice the larger of two numbers is five more than five times the smaller, and the sum of six times the larger and five times the smaller is 265. What is the larger number?
- a.) 44 b.) 40 c.) 30 d.) 27
262. Car A heads north at 45 kph. Exactly 12 minutes after, car B follows at a steady speed of 54 kph. How long does it take for car B to overtake car A?
- a.) 3 b.) 2 c.) 1 d.) 1.5
263. What is the distance between two endpoints, A (-3, 2) and B (1, 5)?
- a.) 0 b.) 3 c.) 4 d.) 5
264. Two cars started from the same point at 4:11 AM. They travelled in opposite directions at 35 mph and 45 mph respectively. At what time will they be 576 miles apart?
- a.) 10:12 AM b.) 10:23 AM c.) 11:23 AM d.) 11:12 AM

265. A man travels 6 miles an hour. After traveling for 4 hours another man starts at the same place, following at the rate of 9 miles an hour. How long will it take for the second man to overtake the first?
- a.) 6 hours b.) 8 hours c.) 9 hours d.) 10 hours
266. If $f(x) = \frac{3x^3}{2}$ and $f(x + 3) = 96$, what is x ?
- a.) 1 b.) 8 c.) 2 d.) 3
267. A pool can be filled by pipe A in 180 minutes and by pipe B in 360 minutes. When the pool is full, it can be drained by pipe C in 720 minutes. If the pool is initially empty and all three pipes are open, how many hours will it take to fill up the pool?
- a.) 2.4 b.) 3.6 c.) 4.2 d.) 2.8
268. Given the arithmetic progression 7, 10, 13, 16, ..., what will be the 26th term?
- a.) 82 b.) 85 c.) 79 d.) 76
269. A coconut tree is 45 years old, and an acacia tree is 27 years old. How many years ago was the coconut tree thrice as old as the acacia tree?
- a.) 3 b.) 9 c.) 18 d.) 27
270. Find the slope of the line from A (2, 5) and B (6, 21)
- a.) 0.25 b.) -4 c.) 4 d.) -0.25
271. If x and y are negative integers where $x > y$, which of the following has the highest value?
- a.) x^3y b.) $y-x$ c.) $-xy^2$ d.) xy^3
272. 21 is 150% of what number?
- a.) 0.7 b.) 14 c.) 31.5 d.) 7.1

273. The expression $\frac{x^2y^4z}{x^5yz^2}$ is equal to?
- a.) $x^{-7}y^5z^{-3}$ b.) $x^7y^{-5}z^3$ c.) $x^3y^{-3}z$ d.) $x^{-3}y^3z^{-1}$
274. The first term of an arithmetic progression is 3 and the common difference is 6. What is the value of the 14th term?
- a.) 81 b.) 87 c.) 93 d.) 75
275. Solve for x in the equation: $4^{2x} = 8^{x+3}$.
- a.) 9 b.) 8 c.) 7 d.) 6
276. It takes Kaye 30 minutes to water the plants in the garden. Marie can water the same garden for 45 minutes. How long will it take if they water than plants together?
- a.) 12 minutes b.) 18 minutes c.) 24 minutes d.) 15 minutes
277. Simplify $\sqrt[6]{125}$.
- a.) 5 b.) $1/5$ c.) $\sqrt{5}$ d.) $1/\sqrt{5}$
278. What is the common difference of an arithmetic sequence whose 26th and 15th terms are 325 and 138 respectively?
- a.) 14 b.) 17 c.) 19 d.) 15
279. A chemist has 600 g of salt solution that is 12% salt. How many grams of 20% solution must be added to obtain a 15% solution of salt?
- a.) 720 b.) 180 c.) 360 d.) 540
280. $8^{-2/3}$ is equal to _____.
- a.) 4 b.) $1/4$ c.) $\sqrt{4}$ d.) $1/\sqrt{4}$
281. Paulo earned 83 in his first two quizzes, 79 for the third, and 85 for the fourth. What should he earn for the fifth quiz to earn an average for 85 for his five quizzes?

- a.) 88 b.) 90 c.) 95 d.) 100

282. A student rides away from his home going to school in an automobile at a rate of 36 miles an hour and walks back fast at a rate of 7.2 miles an hour. The round trip took 2 hours. How far does he ride?

- a.) 18 miles b.) 9 miles c.) 15 miles d.) 12 miles

283. The area of a right triangle is 18. One of its angles is 45° . What is the hypotenuse of the triangle?

- a.) 32 b.) 36 c.) $6\sqrt{2}$ d.) 62

284. How many pounds of milk powder that cost \$5.20 per pound must be mixed with 9 pounds of milk powder that cost \$2.70 per pound to make a mixture that costs \$3.70 per pound?

- a.) 6 b.) 7 c.) 8 d.) 9

285. The first term of an arithmetic progression is 7 and the common difference is -4. What term is -57?

- a.) 17 b.) 16 c.) 15 d.) 14

286. What is the alcohol percentage of the mixture that is mixed from 200 L of 15% alcohol solution with 300 L of 22% alcohol solution?

- a.) 19.2% b.) 22% c.) 15% d.) 18.6%

287. A soft drink factory worker has 400 g of sugar solution that is 9% sugar. How many kilograms of 27% sugar solution must be added to obtain a 24% solution of sugar?

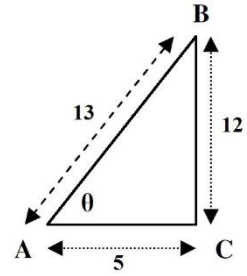
- a.) 0.4 b.) 0.8 c.) 1 d.) 2

288. If Rose saves P15.00 on the first week and adds P6.00 each week, how much will she save on the last week of the fourth month?

- a.) P98.00 b.) P100.00 c.) P105.00 d.) P102.00

289. In the right triangle ABC, what is the value of $\sin \theta$?

- a.) 12/13
- b.) 5/13
- c.) 12/5
- d.) 5/12



290. In the same figure, what is the value of $\tan \theta$?

- a.) 12/13
- b.) 5/13
- c.) 12/5
- d.) 5/12

291. In geometric progression 2, 6, 18, 54, ... how many terms are added if the sum is 19682?

- a.) 6
- b.) 7
- c.) 8
- d.) 9

292. The sum of four consecutive odd integers is 432. What is the largest among those integers?

- a.) 105
- b.) 111
- c.) 113
- d.) 101

293. If a and b are both odd numbers, then which of the following must be an even number?

- a.) $a+b-3$
- b.) ab
- c.) $a+b+2$
- d.) $b+2a$

294. Multiply $(3x + 2y)$ by $(4x - 3y)$

- a.) $12x^2 + 17xy - 6y^2$
- b.) $12x^2 - 17xy - 6y^2$
- c.) $12x^2 + xy - 6y^2$
- d.) $12x^2 - xy - 6y^2$

295. What is the 20th term of an arithmetic sequence whose 33th and 19th terms are 104 and 48 respectively?

- a.) 52
- b.) 54
- c.) 72
- d.) 74

ANSWER KEY:

LANGUAGE PROFICIENCY

1. C , establishing
2. A , nor
3. C , went
4. C
5. E
6. B
7. B , refers
8. B , has
9. C , to solve
10. E
11. C , this
12. D , supposed
13. C , rests
14. C , Von Braun's of NASA
15. C , passers
16. B
17. A
18. B , were
19. B , plan
20. B , play
21. B
22. A , Kung hindi
23. B , amused
24. D , pick
25. C , her
26. D , as careful
27. E
28. B , effect
29. D
30. C , supposed to
31. D
32. B
33. C
34. C
35. A
36. A
37. B

38. A
39. A
40. A
41. A
42. B
43. A
44. A
45. B
46. C
47. B
48. B
49. A
50. C
51. A
52. B
53. C
54. A
55. B
56. B
57. C
58. B
59. A
60. C
61. B , to enjoin means to issue an order or command; to direct or impose with authority
62. C , virago means a bold, impudent, turbulent woman
63. D , to efface means to rub out, erase
64. D , a ravine is a deep narrow canyon
65. D , to vivify means to give or bring life to
66. C , a yearling is a young animal past its first year but not yet two years old
67. A , untimely means happening before the proper time
68. B , zenith means the highest point of any path or course
69. B , an accolade is an award or a special acknowledgement signifying approval or distinction
70. C , to abdicate means to formally relinquish or surrender power, office or responsibility
71. A , compunction means a feeling of uneasiness or regret caused by a sense of guilt, remorse
72. C , rustic means rural; rough; made from rough limbs or trees
73. B , valorous means courageous, valiant
74. D , motif is a recurrent theme or form in an artistic or literary work
75. B , to desiccate means to dry out thoroughly

76. D , relinquish means give up
77. A , oppressive means unreasonably burdensome, heavy
78. A , to rally means to come together for a common purpose
79. C , animosity means bitter, open hostility or enmity; energetic dislike
80. D , to extricate is to release from an entanglement or difficulty
81. B , to accost means to approach and speak to someone, usually in a bold and aggressive manner as with demand
82. D , litigious means inclined to disagree or dispute, especially in lawsuits
83. B , indemnity is a protection from loss or damage; immunity from punishment
84. A , to haggle means to bargain, such as over a price, or dispute in a petty way
85. B , congenial means having a friendly or pleasant disposition, sociable

READING COMPREHENSION

86. B
87. C
88. B
89. B
90. B
91. B
92. D
93. D
94. A
95. A
96. D
97. C
98. B
99. B
100. A
101. A
102. D
103. C
104. D
105. C
106. D
107. C
108. D
109. C
110. D
111. D
112. D
113. A

114.C
115.D
116.D
117.D
118.D
119.B
120.D
121.A
122.D
123.A
124.D
125.D
126.D
127.A
128.D
129.D
130.A
131.B
132.D
133.D
134.A
135.B
136.B
137.D
138.C
139.B
140.C
141.C
142.D
143.D
144.D
145.D
146.C
147.C
148.D
149.A
150.C
151.D
152.C
153.A
154.B
155.C

156. B
157. C
158. B
159. A
160. A
161. C
162. C
163. B
164. A
165. C
166. B
167. A
168. A
169. C
170. D
171. C
172. B
173. D
174. C
175. A

SCIENCE

176. C
177. A , $GPE = mgh$: the gravitational potential energy is directly proportional to height, so as an object is in a high place, it has a high GPE as well
178. A ,
179. C , based on the equation, there are 3 moles of H_3 in 1 mole of N_2 . So $2.5 * 3 = 7.5$ moles of H_3
180. B
181. D
182. C
183. A
184. B
185. C
186. C
187. A , According to Boyle's law, at constant temperature, the volume of a gas is inversely proportional to the amount of pressure exerted on it. If the pressure is doubled, volume will decrease.
188. D , $KE = \frac{1}{2}mv^2 = \frac{1}{2}(500)(3.0)^2 = 2250 J = 2.25 kJ$
189. B , add them with their like terms (in this case, the $\hat{i}$'s and $\hat{j}$'s) $A + B = (4m + 2m)\hat{i} + (3m + (-3m))\hat{j} = 6m\hat{i} + 0m\hat{j} = 6m\hat{i}$

190.B
191.B
192.A
193.A
194.C
195.C, $a = \left \frac{v_f - v_i}{t} \right = \left \frac{50 - 70}{5} \right = \left \frac{-20}{5} \right = 4 \text{ m/s}^2$
196.C
197.A, $F = ma = (2)(0.5) = 1 \text{ N}$
198.C
199.C
200.B
201.C
202.C
203.D
204.D
205.C
206.B, this is the only balanced chemical equation
207.A
208.B
209.A
210.D
211.B
212.C
213.A
214.B
215.A
216.B
217.B
218.D
219.A
220.B
221.B
222.A, Newton's third law of motion (EQUAL and opposite reaction)
223.D
224.D, $F = ma$. The problem states that the plane is moving at a constant speed . It means that it has zero acceleration . Substituting zero to the equation will yield a magnitude of force of zero .
225.A
226.D
227.A, in series circuit, the total resistance is equal to the sum of all resistors. In parallel circuit, total resistance is equal to the sum of reciprocal of each resistor. In the diagram, the two 1Ω resistors together with the unknown resistor R are

connected in parallel. To obtain its resistance, $\frac{1}{1\Omega} + \frac{1}{1\Omega} + \frac{1}{R} = 2\Omega + \frac{1}{R}$. The total resistance of that parallel circuit is in series with the resistors 2Ω and 3Ω . We are given with the total resistance of the circuit as 8Ω . So,

$$\begin{aligned} 8\Omega &= 2\Omega + \left(2\Omega + \frac{1}{R}\right) + 3\Omega \\ 8\Omega &= 7\Omega + \frac{1}{R} \\ 8\Omega - 7\Omega &= \frac{1}{R} \\ 1\Omega &= R \end{aligned}$$

$$228.A, a_{ave} = \left| \frac{v_f - v_i}{t} \right| = \left| \frac{0 - 20}{5} \right| = 4$$

229.C

230.A

231.A

$$232.D, \sqrt{0.5^2 + 1.2^2}$$

233.C

234.D

235.D

MATHEMATICS

236.C

	Today	In 6 years
Ed	x	$x + 6$
Jeb	$x - 14$	$x - 8$

Solve for x

$$\begin{aligned} x + 6 &= 3(x - 8) \\ x + 6 &= 3x - 24 \\ -3x + x &= -24 - 6 \\ -2x &= -30 \\ x &= 15 \end{aligned}$$

Substitute x in Jeb's age today:

$$x - 14 = 15 - 14 = 1$$

In 4 years, **Jeb will be 5 years old.**

More examples: https://youtu.be/eC1_ImKDFNs and <https://youtu.be/L00UVok8Zqo>

Or search in youtube: Age Problems Agham Liknayan

237. **C**

Let x be the larger number; y be the smaller
Solve for the systems of linear equation:

$$3x = 5y + 5 \quad (1)$$

$$6x + 5y = 26 \quad (2)$$

Multiply equation (1) by 2.

$$6x = 10y + 10 \quad (3)$$

Substitute (3) to (2)

$$(10y + 10) + 5y = 26$$

$$y = 17$$

Substitute the value of y to (1) and solve for x

$$x = 30$$

Related link: https://youtu.be/VTk-qhiC_ss

Or search in youtube: Number Word Problems Agham Liknayan

238. **C**

Area of a square is s^2

Making each side to its triple value is $3s$

$$\text{So, } (3s)^2 = 9s^2$$

239. **A**

The rational numbers have similar denominators so combine them by copying the denominator. The numerator cannot simply be added so copy them instead.

240. **B**

Link: https://youtu.be/aTfk6FOb7_4

Or search in youtube: Triangle Analysis Agham Liknayan

241. **A**, divide each monomial of $12x^7y^3 + 8x^6y^4 - 24x^8y^9$ by $2x^4y^2$.

Link: <https://youtu.be/eW5gs6Qfl9Q>

Or search in youtube: Division of polynomials Agham Liknayan

242. **B**, solve for x in $4x = 3x + 5$

243. **A**

Use the formula: $a_n = a_1 + (n - 1)d$

Solve for a_1 and d in the systems of equation:

$$417 = a_1 + (19 - 1)d$$

$$319 = a_1 + (12 - 1)d$$

Related link: <https://youtu.be/EISU3z6VjoA>

Or search in youtube: Arithmetic Sequence Part 3 Agham Liknayan

244. **C**

Related link: <https://youtu.be/wYhGsw8yFow>

Or search in youtube: Arithmetic Sequence Part 1 Agham Liknayan

245. **C**, let $x = 2$ & $y = 1$.

246. **B**, $AE = DE$ & $BE = CE$. Solve for x & y in the systems of equation:

$$3x + 4 = 2y + 5 \text{ (} AE = DE \text{)}$$

$$2x + 1 = y - 3 \text{ (} BE = CE \text{)}$$

247. **A**

Let

$7x$ be the first integer	$7x + 21$ be the fourth integer
$7x + 7$ be the second integer	$7x + 28$ be the fifth integer
$7x + 14$ be the third integer	

Solve for x in the equation:

$$(7x) + (7x + 7) + (7x + 14) + (7x + 21) + (7x + 28) = 1582$$

Finally, substitute x in the first integer, $7x$

Link: <https://youtu.be/cXFJ7bXvZEQ> try also https://youtu.be/VTk-qhiC_ss

Or search in youtube: Number Word Problems Part 4 Agham Liknayan

248. **D**

Let x be the time it takes for Ben to clean the room alone

Solve for x in the equation: $\frac{1}{36} + \frac{1}{x} = \frac{1}{9}$

Related link: <https://youtu.be/Tdl54I7ydql>

Or search in youtube: work problems agham liknayan

249. **B**

Apply the logarithmic property: if $\log_a x = b$ then $x = a^b$

Link: https://youtu.be/_pc3YmV34rA

Or search in youtube: Logarithmic Function Agham Liknayan

250. **A**

It was stated in the problem that

Volume of rectangular solid = volume of a cube

$$(2)(8)(x) = (4)(4)(4)$$

Solve for x

251. **B**

Let x be the time it takes if they carry the pail together

Related link: <https://youtu.be/Tdl54I7ydgI>

Or search in youtube: work problems agham liknayan

252. **B**

Let

$2x$ be the first integer	$2x + 6$ be the fourth integer
$2x + 2$ be the second integer	$2x + 8$ be the fifth integer
$2x + 4$ be the third integer	

Solve for x in the equation:

$$(2x) + (2x + 2) + (2x + 4) + (2x + 6) + (2x + 8) = 470$$

Link: <https://youtu.be/0dFup4fhxlo> try also https://youtu.be/VTk-qhiC_ss

Or search in youtube: Number Word Problems Part 3 Agham Liknayan

253. **B**, radicand may be simplified to a multiple of a pair of a perfect square and a nonperfect square.

Link: <https://youtu.be/YetHoSdb8Vw>

Or search in youtube: Operations on Radical Numbers Agham Liknayan

254. **B**

Let x be Sam's age today

	Today	In 5 years (+5)
--	-------	-----------------

Sam	x	$x + 5$
Roy	$x + 13$	$x + 18$

Solve for x in:

$$x + 18 + 2(x + 5) = 79$$

More examples: https://youtu.be/eC1_ImKDFNs and <https://youtu.be/L00UVok8Zqo>

Or search in youtube: Age Problems Agham Liknayan

255. D

Area of a rectangle = Length * Width

Area of a square = Side * Side

Let W be the width, L be the length, S be the side of the square

$$W = 3L, W = S = 3L$$

$$\text{Area of the square} = S * S = W * W = W^2 = (3L)^2 = 9L^2$$

$$\text{Area of the rectangle: } L * W = L * 3L = 3L^2$$

$$\text{Ratio: } \frac{\text{Area of the rectangle}}{\text{Area of the square}} = \frac{3L^2}{9L^2} = \frac{1}{3}$$

256. B

$$\text{distance} = \text{speed} * \text{time}$$

Let t be the time it takes for them to meet from 120 kilometers apart

Distance travelled by Ben: $45t$ Distance travelled by Kevin: $35t$

They are 120 kilometers apart, so add their distances

$$45t + 35t = 120$$

Solve for t

Link: <https://youtu.be/fkAgzoOdvK4>

Or search in youtube: Distance Problem Agham Liknayan

257. C

Let x be the time it takes for Paul to finish the job alone

Solve for x in the equation:

$$\frac{1}{6} + \frac{1}{8} + \frac{1}{x} = \frac{1}{3}$$

Link: <https://youtu.be/Tdl54I7ydqI>

Or search in youtube: work problems agham liknayan

258. **D**

Let x be Leo's age today

	Today	3 years ago (-3)
Leo	x	$x - 3$
Joy	$x + 9$	$x + 6$

Solve for x in the equation: $(x - 3) + (x + 6) = 29$

Substitute x in $x + 9$

More examples: https://youtu.be/eC1_ImKDFNs and
<https://youtu.be/L00UVok8Zqo>

Or search in youtube: Age Problems Agham Liknayan

259. **A**

Let t be the time it takes

	Today	In t years ($+t$)
Kaye	24	$24 + t$
Pia	9	$9 + t$

Solve for t in the equation: $24 + t = 2(9 + t)$

More examples: https://youtu.be/eC1_ImKDFNs and
<https://youtu.be/L00UVok8Zqo>

Or search in youtube: Age Problems Agham Liknayan

260. **B**

The water has 0% concentration

Solve for x in the equation: $(9000)(0.3) + (x)(0) = (9000 + x)(0.2)$

Related link: <https://youtu.be/k6JIKJKwRHE>

Or search in youtube: Mixture Problem Agham Liknayan

261. **C**

Solve for x & y in the systems of equation:

$$3x = 5y + 5 \text{ and } 6x + 5y = 265$$

Related link: https://youtu.be/VTk-qhiC_ss

Or search in youtube: Number Word Problems Part 1 Agham Liknayan

262. **C**

$Distance = Speed * Time$

Let t be the time for Car B; $t + 12$ be the time for car A

Car	Speed	Time	Distance
-----	-------	------	----------

A	45	$t + 12$	$45(t + 12)$
B	54	t	$54t$

Solve for t in the equation: $45(t + 12) = 54t$

Link: <https://youtu.be/fkAgzoOdvK4>

Or search in youtube: Distance Problem Agham Liknayan

263.D

Use the distance formula: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

264.C

	Speed	Time	Distance
1 st	35	t	$35t$
2 nd	45	t	$45t$

Adding their distances lead to a total of 576 km. Solve for t in the equation:

$$35t + 45t = 576$$

$$t = 7.2 \text{ hrs}$$

Convert 0.2 hrs to minutes

$$0.2 \text{ hrs} \cdot \frac{60 \text{ minutes}}{1 \text{ hr}} = 12 \text{ minutes}$$

So, $t = 7 \text{ hours and } 12 \text{ minutes}$. Add it to 4:11 AM

Link: <https://youtu.be/DZqrm3xnmkw>

Or search in youtube: Distance Problem Agham Liknayan

265.B

Let t be the time it takes for the 2nd man to overtake the 1st man

Man	Speed	Time	Distance
1 st	6	$t + 4$	$6(t + 4)$
2 nd	9	t	$9t$

Equate their distances and solve for t .

$$6(t + 4) = 9t$$

Related link: <https://youtu.be/fkAgzoOdvK4>

Or search in youtube: Distance Problem Agham Liknayan

266.A

$f(x+3) = \frac{3(x+3)^3}{2}$; $f(x+3) = 96$, so, solve for x in the equation:

$$\frac{3(x+3)^3}{2} = 96$$

Link: <https://youtu.be/3jG-3PzGXR5>

Or search in youtube: Composition of Functions Agham Liknayan

267. **A**

$$\frac{1}{A} + \frac{1}{B} + \left(-\frac{1}{C}\right) = \frac{1}{x}$$

Solve for x in the equation:

$$\frac{1}{180} + \frac{1}{360} - \frac{1}{720} = \frac{1}{x}$$

The value of x that obtained here is in terms of *minutes*. Convert the value of x to hours.

Link: <https://youtu.be/TdI54I7ydqI>

Or search in youtube: work problems agham liknayan

268. **A**

Use the formula: $a_n = a_1 + (n-1)d$.

Link: <https://youtu.be/tJInz44Qtgo>

Or search in youtube: Arithmetic Sequence Part 2 Agham Liknayan

269. **C**

Tree	Today	t years ago ($-t$)
Coconut	45	$45 - t$
Acacia	27	$27 - t$

Solve for t in the equation: $45 - t = 3(27 - t)$

More examples: https://youtu.be/eC1_I mKDFNs and

<https://youtu.be/L00UVok8Zqo>

Or search in youtube: Age Problems Agham Liknayan

270. **C**

Use the formula for finding the slope of a line: $m = \frac{y_2 - y_1}{x_2 - x_1}$

Link: <https://youtu.be/HPxcFMSI3F4>

Or search in youtube: Slope of a line Agham Liknayan

271. D , let $x = 2$ & $y = 1$. Substitute in all of the choices
272. B , solve for x in the equation: $1.50x = 21$ Link: https://youtu.be/ypIQ71Fuh6A Or search in youtube: 21 is 150% Agham Liknayan
273. D , eliminate the common in the numerator and denominator. Take note of the property that $\frac{x^a}{x^b} = x^{a-b}$ Link: https://youtu.be/Ti_yX7kjB70 Or search in youtube: Simplifying Rational Expressions Agham Liknayan
274. A , use the formula $a_n = a_1 + (n - 1)d$. Substitute the values from the given and solve for a_n . Related link: https://youtu.be/wYhGsw8yFow Or search in youtube: Arithmetic Sequence Part 1 Agham Liknayan
275. A , this is one example of exponential function. Cancel out the similar bases. It may be done if 4 is substituted by 2^2 and 8 by 2^3 . Finally, solve for x in: $2^{2(2x)} = 2^{3(x+3)}$ Link: https://youtu.be/sg1bdLUyiPA Or search in youtube: Exponential Function Agham Liknayan
276. B Solve for x in the equation: $\frac{1}{30} + \frac{1}{45} = \frac{1}{x}$ Related link: https://youtu.be/Tdl54I7ydqI Or search in youtube: work problems agham liknayan
277. C , $125 = 5^3$, so $\sqrt[6]{125} = \sqrt[6]{5^3} = 5^{\frac{3}{6}} = 5^{\frac{1}{2}} = \sqrt{5}$ Link: https://youtu.be/1qu1N6ANzI0 Or search in youtube: Simplifying Radical Numbers Agham Liknayan
278. B , use the formula $a_n = a_1 + (n - 1)d$ Solve for d in the systems of equation: $325 = a_1 + (19 - 1)d$

$$138 = a_1 + (15 - 1)d$$

Related link: <https://youtu.be/EISU3z6VjoA>

Or search in youtube: Arithmetic Sequence Part 3 Agham Liknayan

279. **C**, solve for x in the equation: $(600)(0.12) + (x)(0.20) = (600 + x)(0.15)$

Related link: <https://youtu.be/k6JIKJKwRHE>

Or search in youtube: Mixture Problem Agham Liknayan

280. **B**

$$8^{-\frac{2}{3}} = \frac{1}{8^{\frac{2}{3}}}; 8 = 2^3; 8^{\frac{2}{3}} = \sqrt[3]{8^2} = \sqrt[3]{(2^3)^2} = \sqrt[3]{2^6} = 2^2; \text{ So, } \frac{1}{8^{\frac{2}{3}}} = \frac{1}{2^2} = \frac{1}{4}$$

Link: <https://youtu.be/md7zZREbRbs>

Or search in youtube: Radical Numbers Agham Liknayan

281. **C**

Let x be his score on the fifth quiz. Solve for x in the equation:

$$\frac{83 + 83 + 79 + 85 + x}{5} = 85$$

282. **D**

Use the formula *distance = speed * time*

Let t be the time while riding; $5t$ be the time while walking (because $36 \div 7.2 = 5$). The round trip took 2 hours, so solve for t in the equation $t + 5t = 2$, $t = \frac{1}{3}$.

Substitute t from the formula,

$$\text{distance} = 36 * \frac{1}{3} = 12$$

Link: <https://youtu.be/ADIJBVWwnaA>

Or search in youtube: Distance Problem Agham Liknayan

283. **C**

$$45^\circ - 45^\circ - 90^\circ$$

$$\text{Area of a triangle} = \frac{1}{2}ab$$

Since $45^\circ - 45^\circ - 90^\circ$ has equal height and base, $a = b$

$$\text{So, area of a triangle} = \frac{1}{2}a^2$$

$$18 = \frac{1}{2}a^2$$

$$a^2 = 36$$

$$a = 6$$

Finally, the hypotenuse is

$$\begin{aligned}h &= \sqrt{6^2 + 6^2} \\h &= \sqrt{72} = \sqrt{36 \cdot 2} \\h &= 6\sqrt{2}\end{aligned}$$

284. **A**, solve for x in the equation: $(x)(5.20) + (9)(2.70) = (x + 9)(3.70)$

Related link: <https://youtu.be/k6JIKJKwRHE>

Or search in youtube: Mixture Problem Agham Liknayan

285. **A**,

Use the formula $a_n = a_1 + (n - 1)d$.

Solve for n in the equation:

$$-57 = 7 + (n - 1)(-4)$$

Link: <https://youtu.be/wYhGsw8yFow>

Or search in youtube: Arithmetic Sequence Part 1 Agham Liknayan

286. **A**

Solve for x in the equation: $(200)(0.15) + (300)(0.22) = (500)(x)$. Multiply the answer by 100.

Related link: <https://youtu.be/k6JIKJKwRHE>

Or search in youtube: Mixture Problem Agham Liknayan

287. **D**

Solve for x in the equation: $(400)(0.09) + (x)(0.27) = (400 + x)(0.24)$. The answer is in terms of grams so convert it into kilograms.

Related link: <https://youtu.be/k6JIKJKwRHE>

Or search in youtube: Mixture Problem Agham Liknayan

288. **C**

Use the formula: $a_n = a_1 + (n - 1)d$

Solve for a_n in the equation: $a_n = 15 + (16 - 1)(6)$

Link: <https://youtu.be/Vi6u0EpWoTk>

Or search in youtube: Arithmetic Sequence Part 4 Agham Liknayan

289. **A**

Link: <https://youtu.be/Owi-jV4yEms>

Or search in youtube: basics of trigonometric functions agham liknayan

290. **C**

Link: <https://youtu.be/Owi-jV4yEms>

Or search in youtube: basics of trigonometric functions agham liknayan

291. **D**, use the formula for geometric progression - $S_n = \frac{a_1(1-r^n)}{1-r}$

$r = \frac{6}{2} = \frac{18}{6} = \frac{54}{18} = 3$; $a_1 = 2$; $S_n = 19682$. Substitute in the equation and solve for n .

$$19682 = \frac{2(1-3^n)}{1-3}$$

$$19682 = \frac{2(1-3^n)}{-2}$$

$$19682 = -(1-3^n)$$

$$-19682 = 1-3^n$$

$$-19682-1 = -3^n$$

$$-19683 = -3^n$$

$$19683 = 3^n$$

$$3^9 = 3^n$$

$$n = 9$$

292. **B**

Let

$2x + 1$ be the first odd integer	$2x + 5$ be the third odd integer
$2x + 3$ be the second odd integer	$2x + 7$ be the fourth odd integer

Solve for x in the equation:

$$(2x + 1) + (2x + 3) + (2x + 5) + (2x + 7) = 432$$

Finally, substitute the value of x in $2x + 7$

Link: <https://youtu.be/q3R6tlrcvA8> try also https://youtu.be/VTk-qhiC_ss

Or search in youtube: Number Word Problems Agham Liknayan

293. **C**, let $a = 4$, $b = 2$

294. **D**, use FOIL method

Link: <https://youtu.be/E-CtMGgqJFE>

Or search in youtube: Multiplication of binomials Agham Liknayan

295. **A**, use the formula $a_n = a_1 + (n - 1)d$

Solve for a_1 and d from the systems of equation:

$$104 = a_1 + (33 - 1)d$$

$$48 = a_1 + (19 - 1)d$$

Substitute those values in a_n : $a_n = a_1 + (20 - 1)d$

Link: <https://youtu.be/EISU3z6VjoA>

Or search in youtube: Arithmetic Sequence Part 3 Agham Liknayan

You may send an email regarding your score in this mock-exam